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Decentralized Federated Learning with Multimodal Prototypes for Heterogeneous Data

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ABSTRACT

Decentralized Federated Learning (DFL) enables collaborative machine learning across numerous devices while avoiding bottlenecks and reliance on a single trusted entity inherent to centralized architectures. However, its practical application is challenged by modern scenarios where data is increasingly multimodal. The key obstacles in such settings are severe data heterogeneity, characterized by non-Independent and Identically Distributed (non-IID) class distributions, and incomplete data, where modalities are often missing across clients. Existing solutions struggle with these challenges, either incurring high communication costs or lacking effective mechanisms for fusing partial information. To overcome these limitations, this work introduces MODALIS, a novel framework for multimodal DFL that achieves superior model performance under data heterogeneity while minimizing network consumption. It pioneers a communication-efficient, prototype-centric protocol in which clients exchange compact, modality-aware class representations rather than high-dimensional model parameters. This process is guided by a multi-objective loss function enforcing inter-modality coherence and representation alignment for effective knowledge fusion. The framework integrates sophisticated architectural innovations, including contextual null embeddings for intelligent data imputation and robust multimodal fusion using adaptive gating and multi-way transformers. The approach is validated through theoretical analysis, providing formal convergence guarantees, and extensive experiments on standard multimodal benchmarks. These results demonstrate that MODALIS achieves superior performance, improving F1 scores by up to 4% under high heterogeneity and reducing communication costs by over 40 times compared to state-of-the-art baselines, establishing it as a highly effective solution for collaborative AI.

1. Introduction

Federated Learning (FL) [1] has emerged as a paradigm-shifting approach that enables collaborative machine learning on distributed datasets without centralizing potentially sensitive user data. This privacy-preserving characteristic is crucial for applications in health-care, finance, and personal devices. However, the conventional server-client topology inherent to most FL systems introduces potential bottlenecks and single points of failure [2]. Decentralized Federated Learning (DFL) [3] mitigates these issues by facilitating direct peer-to-peer communication and model aggregation among clients, thereby enhancing system robustness, scalability, and eliminating reliance on a central trusted entity. Despite these advantages, achieving efficient knowledge aggregation and stable convergence in DFL remains a formidable

challenge, particularly given the multifaceted complexities of real-world data distributions.

Real-world applications rarely rely on a single data source; instead, they typically involve diverse modalities such as images, text, audio, and structured sensor readings. A significant and often overlooked challenge arises from the *multimodality* and *heterogeneity* of data dispersed across participating clients [4]. In decentralized networks, clients frequently contend with *missing modalities* (e.g., a client possesses text data but lacks the corresponding image for a given sample) and highly skewed, non-Independent and Identically Distributed (non-IID) class distributions [5]. Such multifaceted heterogeneity poses a fundamental challenge to existing FL paradigms. Conventional algorithms that rely on parameter averaging are particularly vulnerable; averaging parameters from a client's untrained sub-network, a direct consequence of a missing

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modality, inevitably pollutes the aggregated model and nullifies peer contributions. While more advanced frameworks offer a solution for non-IID data by exchanging compact, class-level feature embeddings, known as prototypes, instead of full model parameters, they were predominantly designed for unimodal tasks and thus lack native mechanisms for multimodal fusion. Finally, the few solutions that address multimodal heterogeneity almost universally rely on a centralized topology for complex operations, such as fusion or imputation, thereby reintroducing the very bottlenecks and single points of failure that DFL is designed to eliminate.

To bridge this critical gap, this paper introduces MODALIS, a novel framework designed for efficient and robust multimodal learning in heterogeneous, decentralized environments. The cornerstone of MODALIS is its *prototype-centric communication protocol*. Instead of transmitting large, high-dimensional model parameters or gradients, clients exchange only low-dimensional *class prototypes*. These prototypes are learnable embeddings that represent the salient, class-discriminative features for specific modalities, as learned by a client from its local data. This strategy achieves substantial communication savings, formalized in [Lemma 1](#), and offers potential privacy benefits by sharing abstract, aggregated representations rather than raw model updates, which can be vulnerable to sophisticated inference attacks [\[6\]](#).

More in detail, this work provides a comprehensive set of technical contributions to advance the state of the art in DFL:

- **Modular framework:** MODALIS is designed and implemented, featuring a novel prototype-based communication protocol at its core. It incorporates several architectural innovations: a Contextual Null Embedding module for intelligent, feature-conditioned imputation of missing modalities; an adaptive gating mechanism that weights modality contributions based on their reliability; and a masked multi-way transformer for robust information fusion that explicitly handles missing-data patterns.
- **Theoretical guarantees:** The work is supported by a rigorous theoretical analysis, which includes a formal convergence proof establishing that MODALIS is guaranteed to reach a stationary neighborhood of an optimal solution ([Theorem 1](#)). Additionally, a detailed lemma quantifies the significant reduction in communication cost of the framework compared to traditional model-passing DFL methods ([Lemma 1](#)).
- **Rigorous validation setup:** An extensive experimental protocol is designed to validate the framework performance. The evaluation is conducted on four diverse and widely recognized multimodal datasets (AVMNIST, CREMA-D, UCI-HAR, CRISIS-MMD). Realistic data heterogeneity is simulated by systematically varying the degree of non-IID class skew (controlled by Dirichlet parameter $\beta \in \{1.0, 0.5, 0.1\}$) and the proportion of missing modalities across clients (controlled by $p_m \in \{0.0, 0.25, 0.5\}$).
- **State-of-the-art results:** The empirical results show that MODALIS consistently outperforms established FL and DFL baselines. Under high-heterogeneity conditions, it achieves a relative F1 score gain of up to $\approx 4.0\%$ over the next-best competing method. The prototype-based communication reduces network overhead by over **40x** compared to methods that exchange full model parameters. Furthermore, its strategy of performing only a single local training epoch ($T = 1$) per communication round minimizes the computational burden on clients and effectively mitigates model drift.

The remainder of this paper is organized as follows: [Section 2](#) reviews pertinent literature in federated and decentralized learning. [Section 3](#) formally defines the problem setup and notation. [Section 4](#) provides a detailed exposition of the MODALIS framework, beginning with its core components and concluding with its operational protocol. [Section 5](#) discusses the theoretical properties of the proposed method. [Section 6](#) presents the comprehensive experimental evaluation. Finally, [Section 8](#) concludes the paper with a summary of findings and a discussion of promising directions for future research.

2. Related work

This section reviews the literature on prototype-based and multimodal FL, as well as on techniques for addressing data heterogeneity, thereby contextualizing the contributions of the work.

2.1. Prototype-based federated learning

Prototype-based FL methods have emerged as a compelling strategy to mitigate high communication costs and improve model robustness against non-IID data. These methods work by exchanging compact, class-representative embeddings (prototypes) rather than voluminous, potentially sensitive model parameters. In the context of centralized FL, the work on FedProto by Tan et al. [\[7\]](#) was seminal. It demonstrated that by having clients exchange only class prototypes with a central server, both communication costs and the negative effects of gradient misalignment under non-IID data could be significantly reduced. The global prototypes shared by the server act as a regularizer, guiding local training. Building on this, FedProc [\[8\]](#) introduced a prototypical contrastive regularizer that forces each client's local feature distributions to align more closely with the globally shared class prototypes, reporting accuracy improvements of $1.6 - 7.9\%$ over standard FedAvg by correcting local training drift on skewed image classification tasks. Similarly, Tran et al. [\[9\]](#) proposed FedNTProto, a personalized FL approach that uses class-specific prototypes to tackle client heterogeneity. By sharing and distilling knowledge through nearest-class prototype representations, each client can retain a personalized model while benefiting from insights into the data distributions of other clients, thereby improving local performance on heterogeneous vision tasks.

The concept of prototype exchange has also been explored in DFL. Qi et al. [\[10\]](#) developed PearFL, a peer-to-peer FL system in which neighboring clients periodically exchange lightweight prototypes rather than full models. This decentralized prototype-sharing mechanism was shown to improve communication efficiency and model consistency across edge devices, with experiments indicating faster convergence and higher accuracy compared to centralized FL in resource-constrained, heterogeneous networks. Li et al. [\[11\]](#) proposed DeProFL, a DFL method tailored for dynamic Internet of Things (IoT) systems that communicate prototype representations. In each round, nodes transmit only learned feature prototypes (e.g., first-layer feature vectors) to their neighbors, enabling the system to handle time-varying topologies by achieving consensus at the prototype level. DeProFL reportedly obtained comparable accuracy (within $0.5 - 5\%$ of centralized methods) while converging faster under non-convex objectives. Zhang et al. [\[12\]](#) combined blockchain technology with prototype learning to eliminate the need for a central server in FL. Their framework uses an immutable ledger to synchronize global class prototypes across clients, addressing server-failure vulnerabilities and improving communication efficiency and test accuracy across several non-IID benchmarks.

2.2. Multimodal learning in federated settings

Applying FL to multimodal data introduces unique challenges related to managing diverse data types, handling variations in modality availability across clients, and effectively and securely fusing information. Several centralized approaches have been proposed to tackle these issues. Chen and Zhang [\[13\]](#) proposed FedMBridge, which utilizes a topology-aware hypernetwork to generate “bridge” parameters, thereby enabling knowledge sharing among clients with different model architectures or modalities. Wang et al. [\[14\]](#) addressed multimodal FL by avoiding a unified multimodal model and instead training multiple single-modal models federatedly. Each client contributes to learning modality-specific feature extractors, enabling clients with limited modalities to benefit from others (e.g., in pathology, imaging-only hospitals could contribute to a federated genomics model). Rahman and Nguyen [\[15\]](#) proposed mmFL, which employs model personalization via

meta-learning for clients with heterogeneous modalities. Modality sets cluster clients, and meta-learning adapts a global model to each cluster, reportedly improving accuracy by over 7% compared to vanilla FL. Huang et al. [16] explored various fusion strategies, including early, late, and attention-based, for multimodal data in centralized FL, finding that attention-based fusion was more adaptive but costlier. These approaches, while advancing multimodal FL, primarily operate under a centralized paradigm and may not directly translate to the complexities and constraints of fully decentralized systems.

2.3. Handling data and modality heterogeneity

Addressing data heterogeneity, particularly non-IID distributions and missing modalities, is a critical prerequisite for the successful deployment of practical FL systems. Numerous specialized techniques have been developed, primarily within the centralized FL paradigm. Bao et al. [17] introduced PmcmFL, which employs prototypes as learnable placeholders for missing modalities, formulating a modality-missing loss and an unimodal inference strategy. This method reportedly improved global model performance by 3.7% when 50% of modalities were missing. Poudel et al. [18] augmented client data with similar examples from public datasets for missing modalities in their CAR-MFL method, where clients retrieve matching samples from a public repository to fill missing inputs. Others leverage more advanced machine learning techniques, such as contrastive learning and graph neural networks, to learn meaningful representations for absent modalities by aggregating information from graph neighbors [19]. An et al. [20] presented a two-step aggregation for modality-incomplete FL in healthcare: intra-cluster FL aggregation for modality-specific encoders, then inter-cluster prototype sharing and contrastive integration. Yu et al. [21] proposed FedInMM, incorporating a Long Short-Term Memory (LSTM)-based feature fusion module to handle incomplete modalities by learning temporal patterns of modality availability and dynamically adjusting feature combination. Similarly, Nguyen et al. [22] proposed CreamFL, which addresses modality heterogeneity by clustering clients based on their available modalities and data characteristics. Cluster-specific models are then trained, with a central server coordinating the learning process across the clusters. More recently, Yin et al. [23] proposed FedMAC. This centralized multimodal FL framework combines a federated self-attention fusion module with an adaptive continual update strategy and diffusion-based reconstruction of missing image modalities to improve robustness under non-IID and modality-incomplete data. In parallel, Le et al. [24] introduced MFCPL, a cross-modal prototype-based multimodal FL approach in which a central server maintains complete class prototypes in a shared representation space and uses them to regularize both modality-shared and modality-specific representations, explicitly targeting severely missing-modality settings.

As summarized in Table 1, despite these innovative advances, the vast majority of existing solutions depend on a central server for coordination, data retrieval, or model aggregation. To the best of current knowledge, no prior work comprehensively combines a fully decentralized learning protocol with a multimodal prototype fusion mechanism that is inherently capable of robustly and simultaneously handling both arbitrary missing modality scenarios and severe non-IID data distributions. The MODALIS framework presented herein is designed specifically to fill this important research gap.

3. Problem statement

This work considers a decentralized system composed of N clients, indexed by $i \in \{1, \dots, N\}$. These clients are interconnected through a communication network, which can be formally represented as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. In this graph, $\mathcal{V} = \{1, \dots, N\}$ constitutes the set of clients (vertices), and $\mathcal{E}$ represents the set of available communication links (edges) between pairs of clients. A client i is constrained to exchange information directly only with its immediate neighbors, defined

by the set $\mathcal{N}_i = \{j \mid (i, j) \in \mathcal{E}\}$. The system operates in a fully decentralized manner, meaning there is no central coordinating server to orchestrate communication or aggregate models. This local-neighborhood communication model is standard in decentralized optimization and DFL, as it keeps the per-round communication cost of each client proportional only to its degree in $\mathcal{G}$, rather than to the total number of participants N . Although only immediate neighbors exchange messages at each round, information propagates multi-hop across the network through successive rounds of prototype aggregation, enabling clients to eventually benefit from data residing at distant nodes. The impact of the connectivity pattern and the efficiency of this indirect knowledge sharing are captured in the analysis in Section 5 via the mixing properties of the communication matrix and the parameter ρ .

Each client i possesses a private, local dataset denoted by $\mathcal{D}_i = \{(\mathbf{x}_i^{(s)}, y_i^{(s)}, \mathbf{m}_i^{(s)})\}_{s=1}^{n_i}$, where $n_i = |\mathcal{D}_i|$ is the number of data samples available locally on client i . Each data sample (s) in this dataset consists of three components:

- A multimodal input $\mathbf{x}_i^{(s)} = \{\mathbf{x}_i^{M,(s)}\}_{M \in \mathcal{M}}$, where $\mathcal{M}$ is the global set of all possible modalities present within the system (e.g., $\mathcal{M} = \{\text{image, text, audio}\}$). The term $\mathbf{x}_i^{M,(s)}$ denotes the data pertaining to modality M for sample s .
- A class label $y_i^{(s)} \in \{1, \dots, K\}$, where K represents the total number of distinct classes across the entire federated system.
- A modality availability mask $\mathbf{m}_i^{(s)} = \{m_i^{M,(s)}\}_{M \in \mathcal{M}}$, being a binary vector where $m_i^{M,(s)} = 1$ indicates that modality M is present for sample s on client i , while $m_i^{M,(s)} = 0$ signifies that it is absent.

The system is defined by inherent heterogeneity, which manifests in several critical forms:

- **Data heterogeneity:** The local data distributions $P_i(\mathbf{x}, y, \mathbf{m})$ can vary significantly and arbitrarily across clients. This heterogeneity encompasses non-IID class distributions (label skew), where certain classes may be heavily over- or under-represented across clients, and feature distribution skew, where the statistical properties of features for the same class can differ across clients [2].
- **Modality heterogeneity:** Clients may possess different subsets of the global modality set $\mathcal{M}$. For instance, one client might only have image data, while another might have both image and text data. Furthermore, even for a specific client i , individual modalities can be missing for certain data samples (i.e., $m_i^{M,(s)} = 0$ for some $M \in \mathcal{M}$). The pattern of missing modalities can be arbitrary and differ across both clients and samples.

The primary objective is to enable each client i to collaboratively train a powerful, personalized multimodal model parameterized by θ_i solely by interacting with its direct neighbors in a decentralized fashion. The overarching goal is to minimize a global loss function defined over the union of all local data, expressed as $F(\theta_1, \dots, \theta_N) = \sum_{i=1}^N \frac{n_i}{\sum_j n_j} \mathbb{E}_{(\mathbf{x}, y, \mathbf{m}) \sim \mathcal{D}_i} [\mathcal{L}_{\text{task}}(\theta_i; (\mathbf{x}, y, \mathbf{m}))]$. This optimization must be achieved while strictly respecting the constraints of decentralized communication and ensuring robust performance despite the aforementioned sources of systemic heterogeneity.

3.1. Notation

Table 2 provides a comprehensive summary of the principal notation employed throughout this paper to ensure clarity and consistency.

4. MODALIS Framework

The MODALIS framework provides a complete solution for decentralized multimodal learning. This section details its core elements using a top-down approach. It first outlines the high-level, round-based operational protocol that governs collaborative training. Subsequently, it

Table 1

Comparative analysis of state-of-the-art FL methods.

Method	Year	Topology	Communication Type	Multimodal	Missing Modality	Non-IID	Personalization	Domain
FedProto [7]	2022	Centralized	Prototypes	✓	✓	✓	✗	Image classification
FedProc [8]	2023	Centralized	Prototypes	✓	✓	✓	✗	Image classification
FedNTProto [9]	2024	Centralized	Prototypes	✓	✓	✓	✓	Vision tasks
DeProFL [11]	2024	Decentralized	Prototypes	✓	✓	✓	✗	IoT systems
PearFL [10]	2025	Decentralized	Prototypes	✓	✓	✓	✗	Edge networks
DFPL [12]	2025	Blockchain-based	Prototypes	✓	✓	✓	✗	Non-IID benchmarks
FedMBrige [13]	2024	Centralized	Parameters	✓	✓	✓	✗	Heterogeneous arch.
FedMMR [14]	2024	Centralized	Parameters	✓	✓	✓	✗	Missing-modality recon.
mmFL [15]	2024	Centralized	Parameters	✓	✓	✓	✓	Modality clustering
Huang et al. [16]	2024	Centralized	Parameters	✓	✓	✓	✗	Fusion strategies
PmcmFL [17]	2023	Centralized	Parameters	✓	✓	✓	✗	Prototype masking
CreamFL [22]	2023	Centralized	Parameters	✓	✓	✓	✗	Contrastive ensemble
CAR-MFL [18]	2024	Centralized	Parameters	✓	✓	✓	✗	Public-data augmentation
MIFL [19]	2024	Centralized	Parameters	✓	✓	✓	✗	GNN-based imputation
An et al. [20]	2024	Centralized	Parameters	✓	✓	✓	✗	Healthcare encoders
FedInMM [21]	2024	Centralized	Parameters	✓	✓	✓	✗	Temporal fusion
FedMAC [23]	2025	Centralized	Parameters	✓	✓	✓	✗	Self-attention fusion
MFCPL [24]	2025	Centralized	Prototypes	✓	✓	✓	✗	Severely missing modality
MODALIS (our work)	2025	Decentralized	Prototypes	✓	✓	✓	✓	General multimodal

provides a detailed exposition of the key components that enable this protocol: the client's internal architecture, the mechanisms for prototype management, and the multi-objective loss function.

4.1. High-level protocol and operational flow

MODALIS operates through a dynamic, round-based asynchronous protocol. This protocol defines the interactions between clients and their internal computational flow, orchestrating the static components described in subsequent sections. The overall process for a client i during a single round t consists of five main steps, as illustrated in Fig. 1:

- 1. Receive prototypes:** At the start of round t , client i processes any new prototype libraries $\{\mathbf{P}_i^j\}$ that have arrived from its neighbors $j \in \mathcal{N}_i$ since its last aggregation phase. The notation i_j signifies the round in which neighbor j 's prototype was generated, accommodating stragglers and variable network latencies inherent to asynchronous DFL. The protocol does not require all neighbors to report in lockstep; instead, each client aggregates the most recent information available to it at the time of its local update.
- 2. Aggregate prototypes:** Client i computes an aggregated prototype library $\mathbf{P}_i^t$. This aggregation combines its local prototypes from the previous round, $\mathbf{P}_i^{t-1}$, with the received neighbor prototypes using the count-weighted averaging scheme from Eq. (2).
- 3. Local training:** Client i performs exactly one epoch ($T = 1$) of training over its local dataset $\mathcal{D}_i$. During this epoch, it utilizes the freshly computed aggregated prototypes $\mathbf{P}_i^t$ to calculate auxiliary loss components (detailed in Section 4.4) and updates its local model parameters θ_i . The choice of $T = 1$ is deliberate to minimize local computation and mitigate client drift under significant data heterogeneity [25].
- 4. Update local prototypes:** After the local training epoch, client i updates its local prototype library from $\mathbf{P}_i^{t-1}$ to $\mathbf{P}_i^t$, using accumulated feature embeddings and the adaptive EMA mechanism (detailed in §4.3). This update allows local prototypes to evolve based on newly observed data while being stabilized by their recent history.
- 5. Send prototypes:** Client i transmits its new local prototype library $\mathbf{P}_i^t$ to all its direct neighbors $j \in \mathcal{N}_i$, which disseminates its latest learned class representations to its peers for use in the next round of aggregation.

These five steps, comprising a full communication round, are formally detailed in Algorithm 1. This algorithm encapsulates the entire operational logic of a single client, from aggregation to communication.

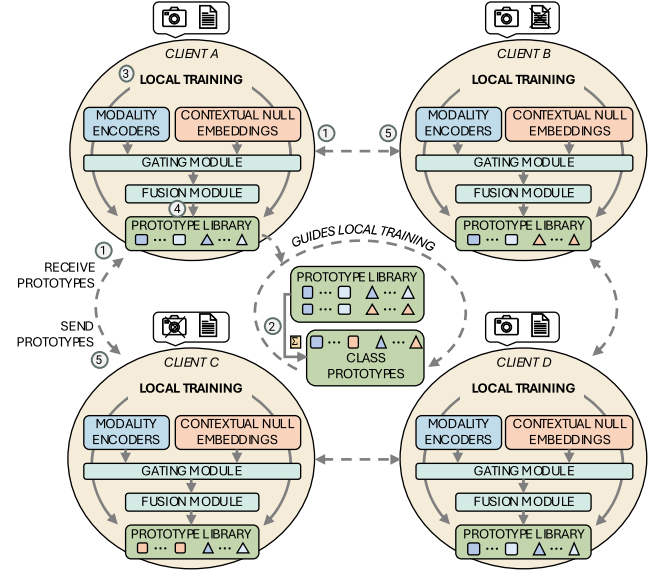


Fig. 1. MODALIS operational flow in a decentralized architecture.

4.2. Client system architecture

The local training phase of the protocol (Step 3) is executed by a sophisticated multimodal neural network, parameterized by θ_i , which is specifically designed to process multimodal inputs, handle missing data, and generate representations for both task performance and prototype generation. As illustrated in Fig. 2, the client's neural network defines the forward pass for a given sample s . Its components are executed sequentially to produce the embeddings and logits required for both final classification and subsequent multi-objective loss calculation. The key architectural components are:

- 1. Modality encoders ($\mathcal{E}_M$):** For each modality $M \in \mathcal{M}$, a dedicated encoder network $\mathcal{E}_M$ maps the raw input data $\mathbf{x}_i^{M,(s)}$ to a fixed-dimensional feature embedding $\mathbf{z}_i^{M,(s)} \in \mathbb{R}^d$. The architecture of $\mathcal{E}_M$ is chosen to be appropriate for the data type of modality M (e.g., a CNN for images, a Transformer for text).
- 2. Contextual null embedding ($\mathcal{N}_M$):** This module is a cornerstone for handling missing modalities. If modality M is absent for a sample s (i.e., $m_i^{M,(s)} = 0$), the $\mathcal{N}_M$ module generates a substitute embedding $\mathbf{z}_i^{M,(s)}$. It utilizes a learnable base null embedding vector $\mathbf{e}_{\text{null_base}}^M \in \mathbb{R}^d$.

Table 2

Summary of key notation.

Symbol	Description
B	Local mini-batch size
β	Dirichlet distribution parameter for simulating non-IID label skew
C	Final classification head network
d	Dimensionality of embedding spaces
D_i	Local dataset of client i
δ_p^2	Bound on expected squared prototype drift per round (Assumption 5)
$\mathcal{E}_M$	Encoder network for modality M
$\mathbf{e}_i^{(s)}$	Fused multimodal embedding for sample s on client i
η	Learning rate
F	Multimodal fusion module
G^2	Bound on expected squared norm of stochastic gradients (Assumption 3)
$\mathcal{G}$	Gating module for modality embeddings
$\mathcal{G}, \mathcal{V}, \mathcal{E}$	Communication graph, vertices (clients), edges (links)
i, j	Client indices
K	Total number of classes
k	Class index, $k \in \{1, \dots, K\}$
L	Smoothness constant for local objective functions (Assumption 1)
$\mathcal{L}$	Loss function component
M	Index representing a specific modality
$\mathcal{M}$	Set of all possible modalities in the system
$m_i^{M,(s)}$	Binary mask: 1 if modality M is present for sample s on client i , 0 otherwise
m_t	EMA momentum parameter for prototype update at round t
N	Total number of clients
$\mathcal{N}_i$	Set of neighbors of client i in graph $\mathcal{G}$
n_i	Number of samples in local dataset D_i
$\mathcal{N}_M$	Contextual Null Embedding module for modality M
θ_i^t	Model parameters of client i at the end of round t
$\mathbf{P}_i^t$	Local prototype library of client i at the end of round t
$\bar{\mathbf{P}}_i^t$	Aggregated prototype library computed by client i at round t
p_m	Proportion parameter for simulating missing modalities
P_1, P_2	Projection head networks
$\mathbf{p}_i^t(*, k)$	Local prototype vector of client i for class k , type $*$ (modality/fusion), at round t
$\bar{\mathbf{p}}_i^t(*, k)$	Aggregated prototype vector computed by client i for class k , type $*$, at round t
T	Number of local training epochs per round (set to $T = 1$)
$\mathbf{r}_i^{(s)}$	Projection of fused embedding $\mathbf{e}_i^{(s)}$ via P_1 for sample s on client i
ρ	Spectral gap related parameter of the network mixing matrix (Assumption 4)
σ^2	Bound on variance of stochastic gradients (Assumption 2)
s_{DCL}	Scaling factor for $\mathcal{L}_{DCL}$
t	Communication round index
R	Total number of communication rounds
τ_{DCL}	Temperature parameter for $\mathcal{L}_{DCL}$
w	Weight hyperparameter for a loss component
$(\mathbf{x}_i^{(s)}, \mathbf{y}_i^{(s)}, \mathbf{m}_i^{(s)})$	s -th data sample on client i (multimodal input, label, modality mask)
$\mathbf{z}_i^{M,(s)}$	Feature embedding for modality M from sample s on client i
$\hat{\mathbf{z}}_i^{M,(s)}$	Gated feature embedding for modality M from sample s on client i
$\mathbf{z}_i^{tM,(s)}$	Projection of modality embedding $\mathbf{z}_i^{M,(s)}$ via P_2

$\mathbb{R}^d$, which is specific to modality M . Critically, if another modality M' is present for the same sample, $\mathcal{N}_M$ conditions the generation of the null embedding on the feature representation $\mathbf{z}_i^{M',(s)}$ from the available modality. This contextual generation is achieved via an MLP:

$$\mathcal{N}_M(\mathbf{z}_i^{M',(s)}) = \text{LayerNorm} \left(\text{MLP}_{\text{ctx}}(\mathbf{e}_{\text{null_base}}^M; \text{StopGrad}(\mathbf{z}_i^{M',(s)})) + \mathbf{e}_{\text{null_base}}^M \right) \quad (1)$$

where $[\cdot; \cdot]$ denotes concatenation. The use of $\text{StopGrad}(\cdot)$ is crucial as it detaches the context embedding $\mathbf{z}_i^{M',(s)}$ from the computational graph of the imputation module. It ensures that the encoder $\mathcal{E}_{M'}$ is optimized solely for extracting task-relevant features from its

Algorithm 1: MODALIS: client operation.

Input: Client i ; Current model θ_i^{t-1} ; Local dataset D_i ; Previous local prototypes $\mathbf{P}_i^{t-1}$; Neighbor prototypes $\{\mathbf{P}_j^{t-1}\}_{j \in \mathcal{N}_i}$; Hyperparameters $\{w, \eta, T = 1, B\}$.

Output: Updated model parameters θ_i^t ; Updated local prototypes $\mathbf{P}_i^t$.

Initialize new aggregated prototype library $\bar{\mathbf{P}}_i^t$;

foreach type $*$ **and** class k **do**

 Compute $\bar{\mathbf{p}}_i^t(*, k)$ and its count $\bar{c}_i^t(*, k)$ using Eq. (2);

 Store L2-normalized $\bar{\mathbf{p}}_i^t(*, k)$ in $\bar{\mathbf{P}}_i^t$;

$\theta_i \leftarrow \theta_i^{t-1}$ ▷ Load current model parameters Initialize feature accumulators: $\Sigma_k^* \leftarrow \mathbf{0}, \mathbf{v}_k^* \leftarrow \mathbf{0}$ for all types $*$ and classes k ;

for batch_idx $\leftarrow 1$ **to** $\lceil n_i/B \rceil$ **do**

 Sample mini-batch $B_s = \{(\mathbf{x}_s, \mathbf{y}_s, \mathbf{m}_s)\}_{s=1}^{|B_s|}$ from D_i ;

foreach sample $s \in B_s$ **do**

foreach modality $M \in \mathcal{M}$ **do**

if $m_s^M = 1$ **then**

$\mathbf{z}_s^M \leftarrow \mathcal{E}_M(\mathbf{x}_s^M; \theta_i^{t-1})$;

else

 Select context modality M' (if available);

$\mathbf{z}_s^{M'} \leftarrow \mathcal{N}_{M'}(\text{context from } \mathbf{z}_s^{M'} \text{ or null basis}; \theta_i^{t-1})$;

$\{\hat{\mathbf{z}}_s^M\}_M \leftarrow \mathcal{G}(\{\mathbf{z}_s^M\}_M; \theta_i^{t-1})$;

$\mathbf{e}_s \leftarrow F(\{\hat{\mathbf{z}}_s^M\}_M, \mathbf{m}_s; \theta_i^{t-1})$;

$\mathbf{r}_s \leftarrow P_1(\mathbf{e}_s; \theta_i^{t-1})$ and $\{\mathbf{z}_s^{tM}\}_M \leftarrow P_2(\{\mathbf{z}_s^M\}_M; \theta_i^{t-1})$;

$\mathbf{z}_{\text{cls},s} \leftarrow C(\mathbf{e}_s; \theta_i^{t-1})$;

$L_{\text{total}} \leftarrow \sum w_{\text{loss}} \mathcal{L}_{\text{loss}}$ using Eqs. (5)-(9) on batch B_s and prototypes $\bar{\mathbf{P}}_i^t$;

$\theta_i \leftarrow \theta_i - \eta \nabla_{\theta_i} L_{\text{total}}$;

 Update Σ_k^* and $\mathbf{v}_k^*$ using detached embeddings from genuine modalities in B_s ;

$\theta_i^t \leftarrow \theta_i$;

Initialize new local prototype library $\mathbf{P}_i^t$;

foreach type $*$ **and** class k **do**

if $\mathbf{v}_k^* > 0$ **then**

$\bar{\mathbf{z}}_k^* \leftarrow \Sigma_k^* / \mathbf{v}_k^*$;

 Update $\mathbf{p}_i^t(*, k)$ and $\bar{c}_i^t(*, k)$ using EMA as per Eq. (3);

else

$\mathbf{p}_i^t(*, k) \leftarrow \mathbf{p}_i^{t-1}(*, k)$; $\bar{c}_i^t(*, k) \leftarrow \bar{c}_i^{t-1}(*, k)$;

 Store L2-normalized $\mathbf{p}_i^t(*, k)$ and $\bar{c}_i^t(*, k)$ in $\mathbf{P}_i^t$;

return $\theta_i^t, \mathbf{P}_i^t$

modality, preventing its learning objective from being distorted by the auxiliary task of imputing the missing modality M .

- Gating module ($\mathcal{G}$):** This module computes attention-like reliability weights, $\alpha^{M,(s)} \in [0, 1]$, for each modality embedding $\mathbf{z}_i^{M,(s)}$ (including those imputed by $\mathcal{N}_M$). Concretely, a lightweight parametric function receives as input the concatenation of the modality embedding, a one-hot encoding of the modality index, and its availability flag, and produces unnormalized scores that are subsequently normalized with a softmax over the set of modalities present according to $\mathbf{m}_i^{(s)}$. The gated feature embeddings are then calculated as $\hat{\mathbf{z}}_i^{M,(s)} = \alpha^{M,(s)} \cdot \mathbf{z}_i^{M,(s)}$. Because these weights are learned end-to-end jointly with the task and prototype-based losses, modalities whose embeddings are consistently noisy or weakly predictive (e.g., low-quality or miscalibrated sensors) are automatically assigned smaller $\alpha^{M,(s)}$ values during training, so that their contribution to the fused representation is reduced rather than treated as equally reliable.

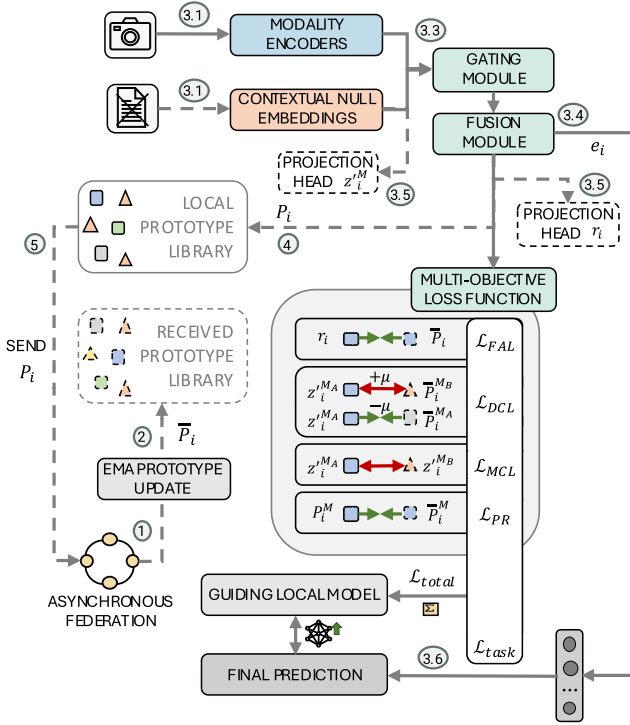


Fig. 2. MODALIS internal architecture.

4. **Multimodal fusion module (F):** A multi-way transformer architecture, inspired by [26], integrates the set of gated modality embeddings $\{z_i^{M(s)}\}_{M \in \mathcal{M}}$ into a single, unified multimodal representation $e_i^{(s)} \in \mathbb{R}^d$. The transformer operates on these gated embeddings as a sequence of modality-specific tokens. It employs an attention mask derived from the modality availability vector $\mathbf{m}_i^{(s)}$, so that tokens corresponding to missing modalities (i.e., those generated purely from contextual null embeddings) neither attend to nor are attended by genuine modalities. Since the fusion module consumes the reliability-weighted embeddings $\hat{z}_i^{M(s)}$, the resulting cross-modal attention effectively treats $\alpha^{M(s)}$ as modality-specific confidence coefficients: modalities that are down-weighted by $\mathcal{G}$ participate only weakly in the attention computation and cannot dominate the fused representation. In this way, the combination of gating and masked multi-way fusion implements an implicit error-correction mechanism that attenuates the impact of noisy or unreliable modalities on $e_i^{(s)}$.
5. **Projection heads (P_1, P_2):** Two MLP-based projection heads map embeddings to specific latent spaces used for auxiliary loss calculations and subsequent prototype generation. Specifically, $\mathbf{r}_i^{(s)} = P_1(e_i^{(s)})$ projects the final fused embedding, and $\mathbf{z}_i'^{M(s)} = P_2(z_i^{M(s)})$ projects the individual pre-gating modality embeddings. These projected representations serve as direct inputs to the loss functions and as features for local prototype updates.
6. **Classifier (C):** A final classification head maps the fused embedding $e_i^{(s)}$ to class logits for computing the task loss.

Together, these architectural features provide a robust solution for handling missing modalities. The combination of contextual imputation ($\mathcal{N}_M$), adaptive weighting ($\mathcal{G}$), and masked fusion (F) enables the model to generate meaningful representations even from incomplete data. The outputs of this architecture, specifically the projected embeddings $\mathbf{r}_i^{(s)}$ and $\{\mathbf{z}_i'^{M(s)}\}_M$, along with the final class logits, serve as the direct inputs for the multi-objective loss function, which is detailed in Section 4.4.

4.3. Prototype management

The prototype exchange, aggregation, and update steps of the protocol (Steps 1, 2, 4, and 5) are governed by the management of class prototypes. These prototypes are the core data structures for knowledge exchange in MODALIS. Clients manage their local prototypes $\mathbf{P}_i$ and compute aggregated prototypes $\bar{\mathbf{P}}_i$ based on information from neighbors. Each prototype, denoted $\mathbf{p}_i^{t,*}(k) \in \mathbb{R}^d$, is an L2-normalized vector representing class k for a specific type $*$. The type $*$ can refer to an individual modality M or the fused representation.

- **Prototype aggregation:** At the beginning of each communication round, a client computes its aggregated prototype library. The aggregated prototype $\bar{\mathbf{p}}_i^{t,*}(k)$ is formed via the count-weighted average specified in Eq. (2). This equation combines the client's local prototype from the previous round ($\mathbf{p}_i^{t-1,*}(k)$) with the most recently received valid prototypes from its neighbors, where $\hat{t}_j$ in the equation denotes the originating round of the neighbor's prototype. This formulation is inherently asynchronous, as $\hat{t}_j$ may be older than $t-1$ for neighbors that are slow to communicate, ensuring the protocol remains robust to network delays. The aggregation is weighted by the number of samples, $c_i^{t-1,*}(k)$, that contributed to each prototype, giving more influence to prototypes built from more data:

$$\bar{\mathbf{p}}_i^{t,*}(k) = \frac{c_i^{t-1,*}(k)\mathbf{p}_i^{t-1,*}(k) + \sum_{j \in \mathcal{N}_i} c_j^{\hat{t}_j,*}(k)\mathbf{p}_j^{\hat{t}_j,*}(k)}{c_i^{t-1,*}(k) + \sum_{j \in \mathcal{N}_i} c_j^{\hat{t}_j,*}(k)} \quad (2)$$

This mechanism is crucial for mitigating non-IID class distributions, as it exposes clients to representations of classes that may be locally scarce. The weights $c_i^{t-1,*}(k)$ and $c_j^{\hat{t}_j,*}(k)$ encode how many training samples contributed to each modality and class-specific prototype, so prototypes from scarce or noisy data have less influence than those supported by many samples. Thus, Eq. (2) performs data-driven, modality-aware reliability weighting, and the aggregated prototypes used as targets in the federated alignment and contrastive losses serve as anchors that pull noisy or underrepresented embeddings toward more reliable regions of the feature space. The resulting vector is L2-normalized.

- **Local prototype update:** After local training, a client updates its local prototypes. It first computes the mean feature embedding $\bar{\mathbf{z}}_k^*$ for each class k and type $*$ using genuine samples observed in the current epoch. The prototype is then updated via an adaptive Exponential Moving Average (EMA):

$$\mathbf{p}_i^{t,*}(k) \leftarrow \text{L2Norm}((1 - m_t) \cdot \text{L2Norm}(\mathbf{p}_i^{t-1,*}(k)) + m_t \cdot \text{L2Norm}(\bar{\mathbf{z}}_k^*)) \quad (3)$$

The adaptive momentum m_t allows for both stability and rapid adaptation to evolving feature distributions.

4.4. Multi-objective loss function

A composite, multi-objective loss function drives local model updates during training. This function, shown in Eq. (4), strategically leverages the aggregated prototypes $\bar{\mathbf{p}}_i^{t,*}(k)$ to balance multiple synergistic objectives and guide the model towards a robust, generalized solution.

$$L_{\text{total}} = w_{\text{task}} \mathcal{L}_{\text{task}} + w_{\text{FAL}} \mathcal{L}_{\text{FAL}} + w_{\text{DCL}} \mathcal{L}_{\text{DCL}} + w_{\text{MCL}} \mathcal{L}_{\text{MCL}} + w_{\text{PR}} \mathcal{L}_{\text{PR}} \quad (4)$$

where $w_{\cdot}$ are scalar balancing hyperparameters. The individual components, computed over a mini-batch B , are defined as:

- **Task loss ($\mathcal{L}_{\text{task}}$):** The primary objective loss for the downstream task is often implemented using a robust loss function like Focal Loss [27] applied to the classifier outputs $\mathbf{z}_{\text{cls}}^{(s)}$ and the ground truth labels

$y_i^{(s)}$ from the current mini-batch. The use of Focal Loss helps address the class imbalance that is often present in individual client datasets.

$$\mathcal{L}_{\text{task}} = \frac{1}{|B|} \sum_{s \in B} \mathcal{L}_{\text{cls}}(\mathbf{z}_{\text{cls}}^{(s)}, y_i^{(s)}) \quad (5)$$

- **Federated alignment loss ($\mathcal{L}_{\text{FAL}}$):** Aligns the projected fused embedding $\mathbf{r}_i^{(s)}$ with its corresponding aggregated fusion prototype, thereby encouraging neighborhood consensus on fused representations:

$$\mathcal{L}_{\text{FAL}} = \frac{1}{|B|} \sum_{s \in B} \left(\left\| \mathbf{r}_i^{(s)} - \text{StopGrad}(\bar{\mathbf{p}}_i^t(\text{fusion}, y_s)) \right\|_2^2 \right) \quad (6)$$

The StopGrad operation prevents gradients from backpropagating into the aggregated prototypes, treating them as fixed targets for alignment during the current local update step.

- **Discriminative contrastive loss ($\mathcal{L}_{\text{DCL}}$):** Enhances intra-class compactness and inter-class separability for modality-specific embeddings using a margin-based contrastive loss against aggregated modality prototypes. Inspired by margin-based classification losses such as ArcFace [28], it can be formulated using a cross-entropy (CE) loss over scaled cosine similarity scores, where an angular margin μ is added to the angle corresponding to the target class prototype. For a genuine modality M in sample s :

$$\mathcal{L}_{\text{DCL}}^{(s,M)} = \text{CE} \left(\text{softmax} \left(\frac{1}{\tau_{\text{DCL}}} \left[s_{\text{DCL}} \cos(\theta_k + \mu \delta_{k,y_s}) \right]_{k=1}^K \right), y_s \right) \quad (7)$$

where θ_k is the angle between $\mathbf{z}_i^{(s)}$ and $\bar{\mathbf{p}}_i^t(M, k)$, μ is an angular margin, and δ_{k,y_s} is the Kronecker delta. The total loss is averaged over all genuine modalities in the batch.

- **Modality coherence loss ($\mathcal{L}_{\text{MCL}}$):** Promotes semantic consistency between representations from different modalities of the same sample by maximizing their cosine similarity:

$$\mathcal{L}_{\text{MCL}} = \frac{1}{|B|} \sum_{s \in B} \sum_{\substack{M_a, M_b \in \mathcal{M}, M_a \neq M_b \\ m_s^{M_a}=1, m_s^{M_b}=1}} \left(1 - \frac{\mathbf{z}_i^{(s)} \cdot \mathbf{z}_i^{(s)}}{\left\| \mathbf{z}_i^{(s)} \right\| \left\| \mathbf{z}_i^{(s)} \right\|} \right) \quad (8)$$

In this sense, it encourages the model to learn shared semantic information across different views of the same underlying concept.

- **Prototype regularization loss ($\mathcal{L}_{\text{PR}}$):** Gently regularizes local prototypes to stay close to the aggregated ones, promoting network-wide stability:

$$\mathcal{L}_{\text{PR}} = \sum_{*,k} \left\| \mathbf{p}_i^{t-1}(*, k) - \text{StopGrad}(\bar{\mathbf{p}}_i^t(*, k)) \right\|_2^2 \quad (9)$$

In summary, the architectural components in Section 4.2 and the losses in Eq. (4) implement an implicit modality-specific error correction scheme. The gating module $\mathcal{G}$ learns reliability weights $\alpha^{M,(s)}$ that downweight noisy modalities, and the masked multiway transformer $\mathcal{F}$ prevents missing or imputed inputs from dominating the fused representation. The prototype-based losses $\mathcal{L}_{\text{FAL}}$ and $\mathcal{L}_{\text{DCL}}$ align modality embeddings with count-weighted aggregated prototypes dominated by well-supported observations. The cross-modality coherence term $\mathcal{L}_{\text{MCL}}$ enforces consistent semantics across modalities, while $\mathcal{L}_{\text{PR}}$ pulls local prototypes toward their neighborhood consensus. As a result, modalities are not fused uniformly but are adaptively weighted according to both learned reliability and the amount of supporting data for each modality and class.

5. Theoretical guarantees

This section presents a theoretical analysis of the convergence properties and communication efficiency of MODALIS, providing foundational guarantees for its behavior in decentralized environments.

5.1. Convergence analysis

The convergence analysis of MODALIS considers a setting with N clients, each optimizing a local objective function $F_i(\theta_i; \bar{\mathbf{P}}_i^t)$ that depends on its local model parameters θ_i and the aggregated prototypes $\bar{\mathbf{P}}_i^t$ available at round t . The global objective implicitly minimizes the average of these local objectives across all clients. Due to the decentralized nature, the use of peer-aggregated prototypes, and the non-convexity of deep learning objectives, the analysis typically aims to show that the algorithm converges to a stationary point or a neighborhood thereof.

The analysis relies on the following standard assumptions:

Assumption 1 (L-smoothness). Each local objective function $F_i(\theta_i; \bar{\mathbf{P}})$ is L -smooth with respect to its model parameters θ , for any fixed set of aggregated prototypes $\bar{\mathbf{P}}$. That is, for all θ, θ' , there exists a constant $L > 0$ such that $\left\| \nabla_{\theta} F_i(\theta; \bar{\mathbf{P}}) - \nabla_{\theta} F_i(\theta'; \bar{\mathbf{P}}) \right\| \leq L \|\theta - \theta'\|$.

Assumption 2 (Unbiased stochastic gradients with bounded variance). The stochastic gradient $\hat{\nabla}_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t)$ computed by client i on a mini-batch of size B at round t is an unbiased estimate of the true local gradient $\nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t)$, i.e., $\mathbb{E}[\hat{\nabla}_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t) | \theta_i, \bar{\mathbf{P}}_i^t] = \nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t)$. Furthermore, the variance of the stochastic gradient is bounded: $\mathbb{E}[\left\| \hat{\nabla}_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t) - \nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t) \right\|^2 | \theta_i, \bar{\mathbf{P}}_i^t] \leq \sigma^2/B$ for some constant $\sigma^2 \geq 0$.

Assumption 3 (Bounded gradients). The expected squared norm of the stochastic gradients is uniformly bounded: $\mathbb{E}[\left\| \hat{\nabla}_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t) \right\|^2] \leq G^2$ for all clients i and rounds t , for some constant $G^2 > 0$. This implies $\mathbb{E}[\left\| \nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i^t) \right\|^2] \leq G^2$.

Assumption 4 (Network connectivity and mixing). The communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is connected. The prototype aggregation process, which averages neighbors' prototypes, is modeled as a stochastic mixing process. Let W_t be a time-varying, row-stochastic matrix representing the communication at round t , where $W_{t,ij} > 0$ only if client i successfully receives a prototype message from client j during that round. Despite asynchronous, potentially delayed communication (i.e., bounded delay), the process is assumed to mix information effectively over time. The rate of convergence towards consensus is related to the spectral properties of the expected mixing matrix $\mathbb{E}[W_t]$, characterized by a parameter $\rho < 1$. Intuitively, better-connected networks (those with larger spectral gaps and smaller diameters) admit smaller values of ρ , leading to faster dissemination of information. In many common random graph models with bounded average degree (e.g., small-world or expander-like graphs), the spectral gap remains bounded away from zero as N grows, so ρ does not approach 1 and prototype information continues to mix efficiently even in large networks. In contrast, very sparse or elongated graphs such as rings or chains have spectral gaps that shrink quickly with N , which corresponds to ρ approaching 1 and a slower decay of the consensus error term in Theorem 1. This dependence on connectivity highlights that the convergence speed of MODALIS is governed by the structural properties of the chosen topology and its effective diameter, even though each round involves only local, neighbor-to-neighbor communication. From a modeling perspective, moderate topology dynamics and temporary link failures can therefore be represented by the time-varying sequence W_t , provided that the union of active links remains connected over sufficiently long windows.

Assumption 5 (Bounded prototype influence and drift). The gradient $\nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i)$ exhibits bounded sensitivity to changes in the aggregated prototypes $\bar{\mathbf{P}}_i$. Specifically, it is assumed that for any θ_i , $\left\| \nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i) - \nabla_{\theta_i} F_i(\theta_i; \bar{\mathbf{P}}_i') \right\| \leq L_p \left\| \bar{\mathbf{P}}_i - \bar{\mathbf{P}}_i' \right\|$ for some Lipschitz constant $L_p > 0$. Furthermore, the expected squared change (drift) in local prototypes per round is bounded, i.e., $\mathbb{E}[\left\| \mathbf{p}_i^t(*, k) - \mathbf{p}_i^{t-1}(*, k) \right\|^2] \leq \delta_p^2$ for all

$i, t, *, k$. The discrepancy between a client's aggregated prototypes $\bar{\mathbf{P}}_i^t$ and a hypothetical global average prototype $\mathbf{P}_{\text{global}}^t$ is bounded, and this bound $E_{\text{proto,disp}}$ is influenced by network mixing properties (quantified by ρ) and the degree of data heterogeneity H across clients.

Theorem 1 (Convergence to a stationary neighborhood). *Under Assumptions 1–5, for MODALIS (Algorithm 1) with $T = 1$ local epoch per round and a constant learning rate η chosen sufficiently small (e.g., $\eta L \leq 1/2$), the average expected squared gradient norm across all clients and rounds converges to a neighborhood of zero:*

$$\frac{1}{R} \sum_{t=0}^{R-1} \frac{1}{N} \sum_{i=1}^N \mathbb{E} \left[\left\| \nabla_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) \right\|^2 \right] \leq \mathcal{O} \left(\frac{F_0 - F^*}{\eta R} + \eta G^2 + \frac{\sigma^2}{B} + \epsilon(\eta, \rho, \delta_p, H, L_p, E_{\text{proto,disp}}) \right) \quad (10)$$

where R is the total number of communication rounds, F_0 is the initial global objective value, F^* is a conceptual minimum, B is the mini-batch size, and $\epsilon(\cdot)$ is a residual error term. This term ϵ captures the aggregate impact of imperfect prototype consensus due to decentralized aggregation (related to ρ and heterogeneity H via $E_{\text{proto,disp}}$), the evolution of the local objectives due to changing prototypes (related to δ_p and L_p), and stochastic noise. As $R \rightarrow \infty$, for sufficiently small η , the algorithm is guaranteed to reach a region where the average expected squared gradient norm is bounded by $\mathcal{O}(\eta G^2 + \sigma^2/B + \epsilon(\cdot))$.

Proof sketch. The proof adapts standard convergence analysis techniques for decentralized non-convex optimization [29,30] and FL with evolving objectives or non-identical data [7,25].

1. Start with the descent lemma for L -smooth functions (Assumption 1) applied to the local objective $F_i(\theta_i^t; \bar{\mathbf{P}}_i^t)$ for client i at round t : $\mathbb{E}[F_i(\theta_i^{t+1}; \bar{\mathbf{P}}_i^t)] \leq F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) - \eta \mathbb{E}[\langle \nabla_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t), \hat{\nabla}_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) \rangle] + \frac{L\eta^2}{2} \mathbb{E}[\left\| \hat{\nabla}_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) \right\|^2]$.
2. Using Assumptions 2 and 3, and for $\eta L \leq 1/2$, this simplifies to: $\mathbb{E}[F_i(\theta_i^{t+1}; \bar{\mathbf{P}}_i^t)] \leq F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) - \frac{\eta}{2} \mathbb{E}[\left\| \nabla_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) \right\|^2] + \frac{\eta\sigma^2}{2B} + \frac{L\eta^2 G^2}{2}$.
3. Account for the change in the objective function due to prototype evolution: $F_i(\theta_i^{t+1}; \bar{\mathbf{P}}_i^{t+1}) - F_i(\theta_i^{t+1}; \bar{\mathbf{P}}_i^t)$. This difference can be bounded using Assumptions 5 and 4, which introduces error terms dependent on L_p, δ_p, ρ , and H .
4. Average the inequality over all N clients. Define a global potential function $\Phi^t = \frac{1}{N} \sum_i F_i(\theta_i^t; \bar{\mathbf{P}}_i^t)$. Sum the averaged inequalities over communication rounds $t = 0, \dots, R-1$.
5. Apply a telescoping sum argument to $\mathbb{E}[\Phi^{t+1}] - \mathbb{E}[\Phi^t]$. Rearrange terms to isolate the average squared gradient norm $\frac{1}{RN} \sum_{t=0}^{R-1} \sum_{i=1}^N \mathbb{E}[\left\| \nabla_{\theta_i} F_i(\theta_i^t; \bar{\mathbf{P}}_i^t) \right\|^2]$.
6. The remaining terms constitute the right-hand side of the theorem. The term $\epsilon(\cdot)$ aggregates errors from stochastic noise, imperfect prototype consensus, and the drift in local objectives. The use of $T = 1$ local epoch simplifies the analysis by avoiding client drift complexities [25].

The theorem indicates that MODALIS converges to a neighborhood of a stationary point, where the size of the neighborhood is influenced by the learning rate, stochastic gradient variance, mini-batch size, and the compounded error term ϵ . $\square$

5.2. Communication efficiency

Lemma 1 (Prototype communication cost reduction). *Exchanging class prototypes in MODALIS yields a significant reduction in communication overhead per round compared to DFL methods that exchange full model parameters. Let $|\theta|$ denote the total number of parameters in the client model architecture, d be the dimension of the prototype embedding vectors, K be*

the total number of distinct classes, and $|\mathcal{M}_{\text{proto}}|$ be the number of distinct prototype types exchanged per class. Assuming a numerical precision p (e.g., 32 bits), the approximate per-round communication cost for a client to send its prototypes to one neighbor is $C_{\text{proto}} = K \cdot |\mathcal{M}_{\text{proto}}| \cdot d \cdot p$ bits. In contrast, standard DFL scenarios that exchange full model parameters incur a cost of $C_{\text{model}} = |\theta| \cdot p$ bits. Given that for typical deep learning models, $K \cdot |\mathcal{M}_{\text{proto}}| \cdot d \ll |\theta|$, MODALIS achieves substantial communication savings.

Proof. In each round, client i transmits its prototype library $\mathbf{P}_i^t$ to its neighbors. This library contains $K \times |\mathcal{M}_{\text{proto}}|$ vectors, each with d values. The total data transmitted to one neighbor is $K \cdot |\mathcal{M}_{\text{proto}}| \cdot d \cdot p$ bits. Conversely, DFL methods exchanging full models transmit the entire parameter vector θ_i of size $|\theta|$, for a cost of $C_{\text{model}} = |\theta| \cdot p$ bits. For a practical example, if $K = 100$, $|\mathcal{M}_{\text{proto}}| = 3$, and $d = 256$, the number of values is $100 \times 3 \times 256 = 76,800$. If the model has $|\theta| = 10^7$ parameters, the cost ratio is $C_{\text{model}}/C_{\text{proto}} = 10^7/76,800 \approx 130$. This reduction of one to three orders of magnitude directly translates to reduced bandwidth requirements and faster training time, especially when communication is the bottleneck. $\square$

5.3. Privacy considerations

Sharing learned class prototypes, rather than raw model parameters or gradients, intuitively offers potential advantages regarding data privacy. Prototypes are abstract, aggregated representations summarizing the characteristic features of entire classes, learned from multiple local data samples. They do not directly expose information about individual data points, as gradients (which are directly tied to specific mini-batches) or model parameters might. Compared to sharing gradients, which is vulnerable to sophisticated inference attacks such as Deep Leakage from Gradients (DLG) [31], prototype sharing is generally considered to present a reduced attack surface [7]. The aggregation process inherent in forming prototypes further obfuscates individual contributions.

However, MODALIS in its base form does not provide formal privacy guarantees, such as those offered by Differential Privacy (DP) [32]. In the standard DP framework, a randomized mechanism $\mathcal{M}$ that releases messages (e.g., prototype libraries) is said to satisfy (ϵ, δ) -DP if, for any pair of neighboring datasets D and D' differing in a single sample and for any measurable set S in the output space, it holds that $\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta$. The current design neither injects calibrated noise into the shared prototypes nor enforces this constraint, and therefore it does not offer any formal (ϵ, δ) -DP guarantee. Potential privacy risks may still arise; for example, through meticulous analysis of sequences of prototypes on clients where only a few samples represent a given class. In such cases, the corresponding class prototype may implicitly encode characteristics closer to those of individual examples. As a consequence, standard membership inference or related privacy attacks remain, in principle, applicable in low-support regimes, even though the use of aggregated prototypes is generally expected to be more resilient than raw gradient sharing in typical settings.

6. Experimentation

This section details the comprehensive empirical evaluation of MODALIS. The experiments were meticulously designed to: (i) assess its overall classification performance against relevant FL baselines across a diverse suite of multimodal datasets; (ii) evaluate its robustness and adaptability under systematically varying degrees of data heterogeneity, specifically non-IID class distributions and different levels of missing modalities; (iii) quantify its communication efficiency and estimate total training time; and (iv) analyze the contribution of its key architectural and algorithmic components through rigorous ablation studies.

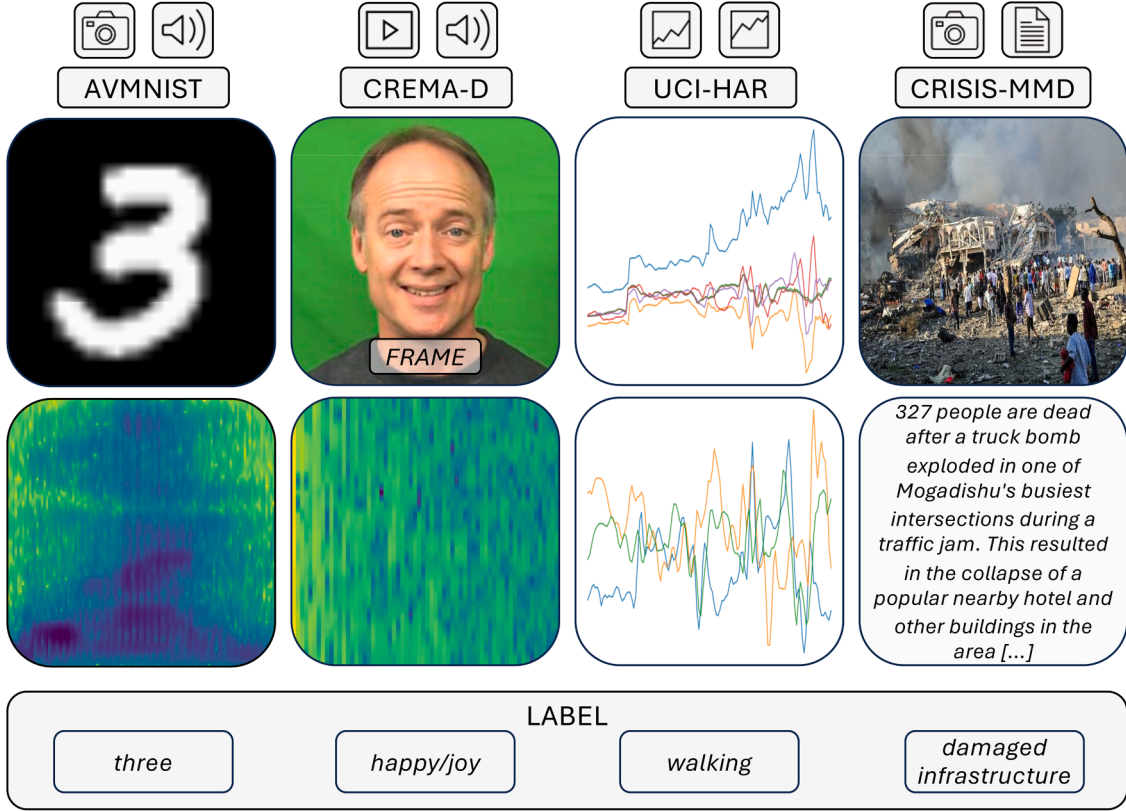


Fig. 3. Example inputs and labels from four multimodal datasets used in this work.

6.1. Experimental setup

- **Datasets:** Four publicly available multimodal datasets were selected to represent a variety of tasks and modality combinations: **AVMNIST** [33] (audio-visual spoken digits), **CREMA-D** [34] (audio-visual emotional speech), **UCI-HAR** [35] (accelerometer and gyroscope data for human activity recognition), and **CRISIS-MMD** [36] (image and text from social media for crisis event classification). These datasets span a spectrum of complexities, from simple digit recognition to nuanced emotional and situational analysis (see Fig. 3).
- **Model architectures:** Modality-specific encoders ($\mathcal{E}_M$) were chosen based on common practices for each data type. The fusion module ($\mathcal{F}$) and other shared components (projection heads P_1, P_2 , classifier C) were kept consistent where possible to ensure fair comparisons.
 - AVMNIST: A simple Convolutional Neural Network (CNN, similar to LeNet-5) for images, and a 3-layer 2D CNN for spectrograms.
 - CREMA-D: A ResNet-18 model [37] processing averaged frames for the video modality, and a 2-layer Bi-directional Long Short-Term Memory network (BiLSTM) operating on MFCC features for the audio modality.
 - UCI-HAR: Two parallel 4-layer 1-D CNNs, each processing the respective 3-axis sensor time series data.
 - CRISIS-MMD: A ResNet-18 model for images, and a Bidirectional Encoder Representations from Transformers (BERT)-base model [38] for text.
- **Heterogeneity simulation:** To mimic challenging real-world federated scenarios, data heterogeneity was introduced artificially across a simulated network of $N = 20$ clients (see Fig. 4 for an illustration of random clients). This heterogeneity has two primary dimensions:
 - **Non-IID class distributions:** The data was partitioned among clients using a Dirichlet distribution $Dir(\beta)$, a standard technique for simulating label skew [39]. Experiments were conducted with

$\beta \in \{1.0, 0.5, 0.1\}$, where a smaller β corresponds to a higher degree of skew.

- **Missing modalities:** The proportion of missing modalities was controlled by a parameter $p_m \in \{0.0, 0.25, 0.5\}$. For a given $p_m > 0$, a fraction p_m of clients received data exclusively from one modality, another fraction p_m received data exclusively from the other, and the remaining $1 - 2p_m$ fraction of clients received complete multimodal data. This setup creates a challenging environment with both fully unimodal and multimodal clients.
- **Baselines and federation topology:** MODALIS was benchmarked against a suite of representative methods: **Centralized** (a theoretical upper bound trained on all data), **FedAvg** [1], **FedProto** [7], **FedInMM** [21], **CreamFL** [22], **PmcmFL** [17], **FedMAC** [23], and **MFCPL** [24]. For a fair comparison with MODALIS, all federated methods were simulated with $N = 20$ participating clients connected via a static Erdős-Rényi (ER) random graph with edge probability $p_{ER} = 0.9$, which provides a realistic but controlled decentralized connectivity pattern for the main reported results. In addition, Section 6.5 includes a dedicated topology-sensitivity analysis in which NEBULA is used to instantiate multiple decentralized graphs to assess the robustness of MODALIS to changes in network structure.

6.2. Implementation details

Training configuration: The framework and all baselines were implemented in PyTorch 2.0 and integrated into NEBULA, an open-source research framework for DFL [40]. NEBULA was selected because it provides a topology-agnostic and model-agnostic environment in which different communication graphs, numbers of clients, and algorithmic variants can be instantiated in a unified and reproducible way. In particular, NEBULA exposes a modular communication layer that allows the same

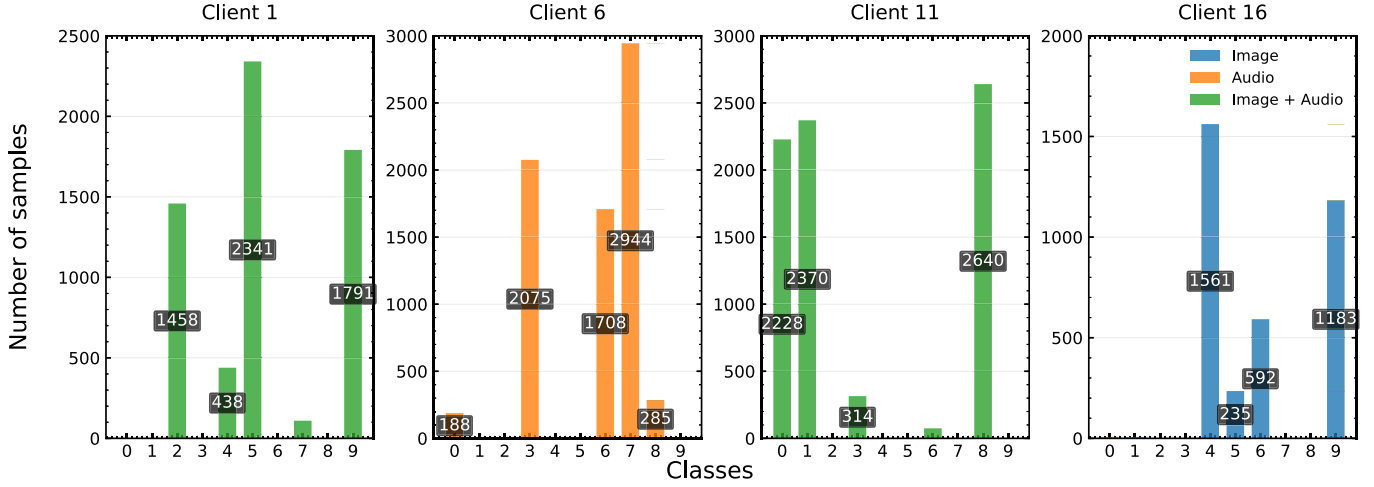


Fig. 4. Four random client data distributions for AVMNIST under specified heterogeneity parameters ($\beta = 0.1$ for class skew, $p_m = 0.25$ for missing modality proportion).

codebase to be executed over a variety of decentralized topologies and custom prototype-exchange protocols, and it supports container-based deployments for running algorithms in realistic multi-process settings. In this work, NEBULA is primarily used in its simulation mode to isolate the algorithmic behavior of MODALIS under controlled heterogeneity and topology configurations, while leaving large-scale containerized deployments with explicit network latencies, client churn, and node mobility to future work. All models were trained using the AdamW optimizer with a learning rate of $\eta = 2 \times 10^{-4}$ and a batch size of $B = 64$. Experiments were conducted for a total of $R = 200$ communication rounds with a common embedding dimension of $d = 128$, being a practical trade-off between representation capacity and communication cost. The framework supports larger embedding dimensions and provides slightly higher accuracy at the expense of proportionally larger prototype messages (see Section 6.5).

Hyperparameter selection for MODALIS: The hyperparameters for MODALIS, particularly the loss weights in Eq. (4), were determined via a grid search on a held-out validation set using the AVMNIST dataset under moderate heterogeneity ($\beta = 0.5$, $p_m = 0.25$). The chosen values, used consistently across all experiments, reflect the relative importance of each objective. The final weights are: $w_{\text{task}} = 1.0$, $w_{\text{FAL}} = 0.5$, $w_{\text{DCL}} = 0.2$, $w_{\text{MCL}} = 0.1$, and $w_{\text{PR}} = 0.01$. This configuration prioritizes the primary task loss, assigns significant weight to the federated alignment loss ($\mathcal{L}_{\text{FAL}}$) to achieve consensus, and uses the remaining terms as auxiliary regularizers to structure the embedding space without overpowering the main objectives. For the $\mathcal{L}_{\text{DCL}}$, the temperature was set to $\tau_{\text{DCL}} = 0.1$, the angular margin to $\mu = 0.3$ radians, and the scale to $s_{\text{DCL}} = 30$, following common practices in representation learning [28]. The adaptive EMA momentum parameters were bounded by $m_{\min} = 0.5$ and $m_{\max} = 0.99$, with the update logic detailed in Algorithm 1.

Evaluation protocol: To ensure robust conclusions, the reported results are the average and standard deviation over three independent runs with different random seeds. The experimental testbed consisted of a compute cluster node equipped with two NVIDIA 3060 GPUs and 128 GB of RAM. Timing estimations assume typical network latencies proportional to data size for the communication phases.

6.3. Performance and convergence comparison

The performance and robustness of MODALIS were evaluated against the baseline suite under two distinct levels of data heterogeneity. First, a moderate-stress scenario with mild label skew ($\beta = 0.5$) was analyzed, followed by a high-stress scenario with severe label skew ($\beta = 0.1$), both of which were combined with increasing proportions of missing

modalities (p_m). Performance was primarily assessed using the Average Macro-F1 Score, a metric chosen for its robustness to the class imbalance created by the non-IID simulation. To provide a holistic evaluation, Communication Cost (in Megabytes per client per round) and Estimated Total Training Time were also reported to quantify efficiency.

6.3.1. Performance under moderate label skew ($\beta = 0.5$)

This setting provides insight into the methods' robustness to missing modalities when label skew is present but less extreme. Table 3 presents the F1 scores and relative drops for this scenario, while Fig. 5 provides a chart visualization of the model performance using $\beta = 0.5$. The results show that MODALIS consistently achieves top-tier performance and remarkable robustness. It either matches or outperforms all federated baselines on AVMNIST, CREMA-D, and CRISIS-MMD across all levels of missing modalities, and remains highly competitive on UCI-HAR, where the MFCPL baseline attains a slightly higher Macro-F1 in several configurations. A key finding is its resilience to performance degradation. For instance, on the UCI-HAR dataset, MODALIS decreases from 92.2% to 84.5% when moving from complete data ($p_m = 0.0$) to maximal missingness ($p_m = 0.5$), which corresponds to a relative loss of roughly 7.7%. Most competing methods lose between 9% and 13% in the same setting. From a dataset-specific perspective, MFCPL emerges as the strongest baseline on UCI-HAR, surpassing MODALIS by 0.5 – 0.6% across $p_m = 0.25$ – 0.5 . This behavior is consistent with its design, which tightly couples class-wise prototypes across modalities and is particularly effective when accelerometer and gyroscope streams are highly correlated.

6.3.2. Performance under high label skew ($\beta = 0.1$)

The high-skew setting represents a significant stress test for any FL algorithm, as it challenges the ability to form coherent global knowledge from highly biased local data. The results in Table 4 show that MODALIS becomes even more critical in this scenario. It achieves the best overall performance across datasets and missing-modality levels, with particularly large margins on AVMNIST and CRISIS-MMD. At the same time, MFCPL is closely matched on UCI-HAR, attaining slightly higher Macro-F1 scores (within 0.5–0.6 points) across all p_m values. However, the relative performance drop of MFCPL with increasing p_m remains slightly larger than Modalis, indicating that the proposed neighborhood-aware prototype regularization is more robust to extreme information loss when averaging across heterogeneous clients. Its advantage over the standard FedAvg was particularly pronounced, with FedAvg collapsing as heterogeneity increased. For example, at $p_m = 0.5$ on AVMNIST, MODALIS achieved a relative F1 score increase of +20.0% over FedAvg. Fig. 6 illustrates the convergence behavior in the most extreme case

Table 3
Performance under moderate label skew ($\beta = 0.5$).

Method	AVMNIST				CREMA-D				UCI-HAR				CRISIS-MMD			
	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$
Centralized	98.5	94.0	89.0	↓9.5%	76.0	69.0	62.1	↓13.9%	95.8	90.0	87.2	↓8.6%	81.0	74.5	67.3	↓13.7%
FedAvg	87.3	81.7	74.2	↓13.1%	59.3	52.7	45.3	↓14.0%	84.2	78.6	70.9	↓13.3%	64.1	57.5	50.1	↓14.0%
FedProto	91.2	86.4	79.3	↓11.9%	64.1	58.6	51.2	↓12.9%	88.1	83.4	76.5	↓11.6%	69.0	62.6	55.9	↓13.1%
FedInMM	92.4	88.7	83.1	↓9.3%	67.2	61.9	55.9	↓11.3%	90.3	86.7	81.2	↓9.1%	72.2	66.0	59.0	↓13.2%
CreamFL	91.8	87.6	81.2	↓10.6%	66.3	60.8	53.7	↓12.6%	89.4	84.9	78.3	↓11.1%	71.1	64.9	57.2	↓13.9%
PmcmFL	93.2	89.5	84.0	↓9.2%	68.1	62.7	56.8	↓11.3%	91.3	87.6	82.1	↓9.2%	73.1	66.8	59.1	↓14.0%
FedMAC	93.0	89.0	83.8	↓9.2%	67.8	62.3	56.2	↓11.6%	91.0	87.2	81.7	↓9.3%	72.8	66.3	53.5	↓19.3%
MFCPL	93.5	89.8	84.7	↓8.8%	68.5	63.0	58.9	↓9.6%	93.1	88.7	83.9	↓9.2%	73.5	67.0	58.3	↓15.2%
MODALIS	94.1	90.1	86.3	↓7.8%	69.0	62.1	60.1	↓8.9%	92.2	88.2	84.5	↓7.7%	74.0	67.4	59.1	↓14.9%

[†] represents the performance decrease from $p_m = 0.0$ to $p_m = 0.5$. Centralized results are from a single run.

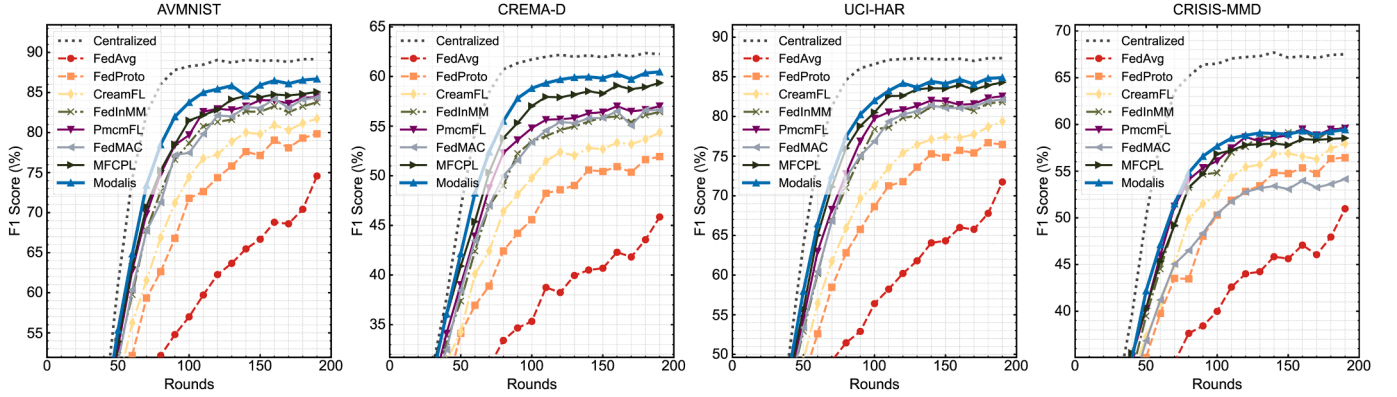


Fig. 5. Model performance (Avg. Macro-F1 Score %) under moderate Non-IID Skew ($\beta = 0.5$) and 50% missing modalities ($p_m = 0.5$).

Table 4
Performance under high label skew ($\beta = 0.1$).

Method	AVMNIST				CREMA-D				UCI-HAR				CRISIS-MMD			
	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$	$p_m=0.0$	$p_m=0.25$	$p_m=0.5$	$(0 \rightarrow 0.5)^\dagger$
Centralized	97.6	92.8	87.9	↓9.7%	74.5	67.5	60.8	↓13.7%	94.7	88.5	81.8	↓12.9%	79.5	72.6	65.7	↓13.8%
FedAvg	86.5	79.2	69.5	↓17.0%	57.2	50.1	42.8	↓14.4%	81.5	74.3	65.7	↓15.8%	61.3	54.1	47.2	↓14.1%
FedProto	90.3	84.1	74.8	↓15.5%	62.1	56.3	48.7	↓13.4%	86.2	79.1	71.3	↓14.9%	66.2	59.3	53.8	↓12.4%
FedInMM	92.8	86.9	79.7	↓13.1%	64.8	58.4	52.6	↓12.2%	88.8	81.9	75.8	↓13.0%	68.7	61.7	58.1	↓10.6%
CreamFL	92.3	85.8	78.3	↓14.0%	64.1	57.2	51.1	↓13.0%	88.1	80.3	73.2	↓14.9%	68.1	60.7	56.7	↓11.4%
PmcmFL	93.4	87.5	80.3	↓13.1%	65.3	58.9	54.1	↓11.2%	88.9	82.5	76.3	↓12.6%	67.3	60.1	54.2	↓13.1%
FedMAC	93.1	87.0	79.9	↓13.2%	65.9	58.6	53.5	↓12.4%	88.8	82.2	76.1	↓12.7%	67.2	59.2	55.7	↓11.5%
MFCPL	93.0	87.2	80.0	↓13.0%	65.0	58.7	53.8	↓11.2%	89.5	83.7	79.9	↓9.6%	68.5	61.5	57.8	↓10.7%
MODALIS	94.3	88.1	83.4	↓10.9%	66.2	59.5	55.1	↓11.1%	88.9	83.2	79.3	↓9.6%	70.2	61.7	60.0	↓10.2%

[†] represents the performance decrease from $p_m = 0.0$ to $p_m = 0.5$. Centralized results are from a single run.

($p_m = 0.5$), showing the learning dynamics under severe data fragmentation.

In this high-heterogeneity regime ($\beta = 0.1$ with p_m up to 0.5), the centralized upper bound already exhibits notable degradation, with Macro-F1 reductions between 9.7% and 13.8% when moving from $p_m = 0.0$ to $p_m = 0.5$ in Table 4. Under the same conditions, all federated methods incur additional losses compared to milder settings, and MODALIS is no exception. For AVMNIST, for example, MODALIS decreases from 94.3% at $p_m = 0.0$ to 83.4% at $p_m = 0.5$, corresponding to a relative drop of 10.9%, whereas FedAvg and FedProto degrade by 17.0% and 15.5%, respectively, over the same range. Similar behavior is observed on CREMA-D, UCI-HAR, and CRISIS-MMD, where MODALIS consistently shows the smallest relative performance loss among all compared methods. These results indicate that the prototype-based multimodal design substantially improves robustness to strong heterogeneity, while not eliminating the inherent performance penalties associated with extreme information loss.

6.3.3. Cross-modal knowledge transfer on unimodal clients

To directly assess the ability of different frameworks to transfer useful knowledge across modalities, the performance of client groups that possessed only a single modality was analyzed in the most challenging setting ($p_m = 0.5, \beta = 0.1$). Table 5 reports the F1 scores for these spe-

cific unimodal clients. The results demonstrate the strong capacity of MODALIS for cross-modal knowledge transfer. On AVMNIST and CRISIS-MMD, unimodal clients using MODALIS achieve the highest F1 scores across all methods, and on CREMA-D, they are essentially tied with the best-performing baseline. For instance, on the AVMNIST dataset, MODALIS image-only clients achieved an F1 score of 82.1%, representing a 19.9% relative gain over FedAvg and outperforming all other advanced baselines. On UCI-HAR, MFCPL slightly surpasses MODALIS for both accelerometer-only and gyroscope-only clients, reflecting the effectiveness of its class-wise prototype matching in this particular sensor configuration.

6.3.4. Cross-dataset transferability

While the primary experiments train and evaluate MODALIS on each dataset separately, it is also important to understand how well a configuration tuned in one environment transfers to other, dissimilar tasks. In the present setup, the datasets differ not only in distribution but also in label spaces and, in some cases, modality semantics (e.g., spoken digits versus social-media crisis posts). As a result, fully zero-shot deployment of a model trained on AVMNIST directly to CRISIS-MMD, without retraining any parameters, is not well-defined: the class sets, loss functions, and input domains are incompatible. Instead, a practically rele-

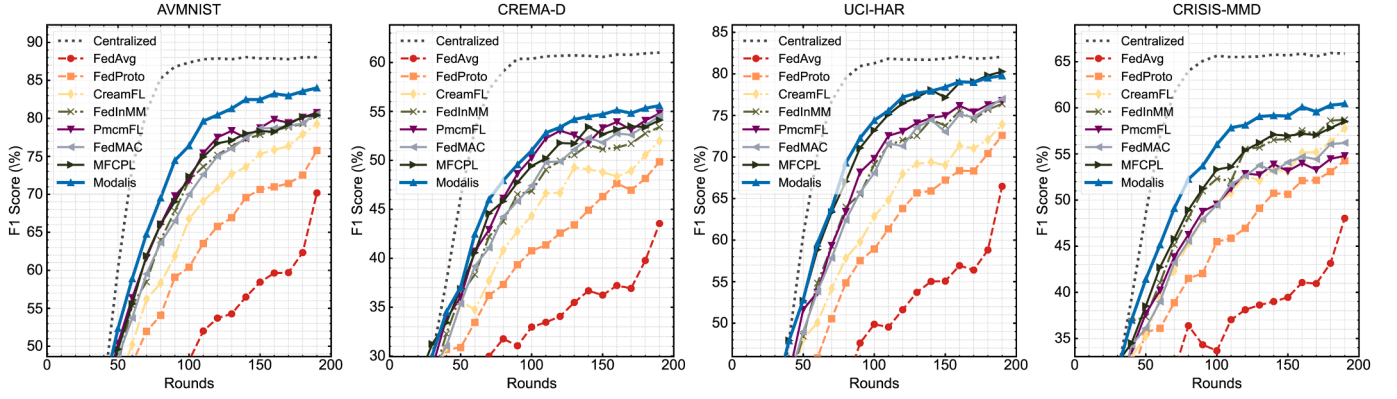


Fig. 6. Model performance (Avg. Macro-F1 Score %) under high Non-IID skew ($\beta = 0.1$) and 50% missing modalities ($p_m = 0.5$).

Table 5

Performance on client subgroups possessing only a single modality, evaluated under the high heterogeneity setting ($p_m = 0.5$, $\beta = 0.1$).

Method ^a	AVMNIST		CREMA-D		UCI-HAR		CRISIS-MMD	
	Image	Audio	Image	Audio	Accelerometer	Gyroscope	Text	Image
FedAvg	68.5±1.0	67.2±1.1	40.1±1.4	38.7±1.5	64.3±1.2	62.9±1.3	45.8±1.6	43.9±1.7
FedProto	73.1±0.8	71.5±0.9	46.2±1.1	44.9±1.2	69.8±0.9	68.3±1.0	51.9±1.3	50.0±1.4
FedInMM	78.3±0.7	76.8±0.8	50.2±1.0	48.8±1.1	74.3±0.8	73.0±0.9	56.3±1.1	54.5±1.2
CreamFL	76.5±0.9	74.9±1.0	48.1±1.2	46.7±1.3	71.6±1.0	70.2±1.1	53.6±1.4	51.7±1.5
PmcmFL	79.0±0.6	77.4±0.7	51.5±0.9	50.1±1.0	75.0±0.7	73.6±0.8	57.5±1.0	55.6±1.1
FedMAC	78.8±0.6	77.0±0.7	51.0±0.9	49.5±1.0	74.8±0.7	73.3±0.8	57.0±1.0	55.3±1.1
MFCPL	80.0±0.6	78.5±0.7	52.8±0.8	51.5±0.9	78.6±0.6	77.3±0.7	58.0±0.9	56.1±1.0
MODALIS	82.1±0.5	80.5±0.6	53.0±0.9	51.7±1.0	78.1±0.6	76.8±0.7	58.9±1.0	57.0±1.1

^a These clients use only their available modality for inference, but their models are trained within the respective FL/DFL frameworks.

vant notion of cross-dataset transferability is considered, in which the algorithmic configuration (hyperparameters and architectural choices) is tuned on a source dataset and then reused without modification when training on other target datasets.

To this end, two settings are compared for MODALIS, FedAvg, and the multimodal baseline MFCPL on CREMA-D, UCI-HAR, and CRISIS-MMD under moderate heterogeneity ($\beta = 0.5$, $p_m = 0.25$): (i) an oracle per-dataset tuning, where a small grid search is allowed on each target dataset; and (ii) an AVMNIST-tuned configuration, in which all hyperparameters are fixed to the values selected on AVMNIST and are not re-optimized on the target. The results, summarized in Table 6, show that MODALIS incurs only a modest performance degradation when transferring its AVMNIST-tuned configuration: on CREMA-D, UCI-HAR, and CRISIS-MMD, the Macro-F1 decreases by approximately 1.9, 1.2, and 1.9 points, respectively, while still attaining the best absolute scores across all methods. By contrast, FedAvg shows larger relative drops of 4.1% – 6.1% under the same protocol. MFCPL occupies an intermediate position: it clearly improves over FedAvg in both tuned and transferred configurations, but its transfer-induced degradation remains noticeably higher than that of MODALIS, with decreases of about 3.6, 2.0, and 3.5 points on CREMA-D, UCI-HAR, and CRISIS-MMD, respectively. These findings indicate that some dataset-specific tuning remains beneficial in practice, but MODALIS is the most robust to cross-dataset hyperparameter transfer. A single configuration tuned on AVMNIST yields near-oracle performance on other tasks, whereas both FedAvg and MFCPL are more sensitive to per-dataset tuning.

6.4. Resource cost and training time analysis

As established theoretically in Lemma 1, MODALIS achieves substantial communication efficiency by exchanging low-dimensional prototypes rather than full model parameters. This section complements that analysis with a quantitative study of the overall resource footprint, including computation time and node-level CPU, GPU, and RAM

utilization (see Table 7). The results show that a MODALIS client transmits approximately 0.11 MB per round, which is more than 40× smaller than communication-heavy methods such as FedAvg and PmcmFL (each sending around 4.75–4.80 MB) and is still substantially below advanced multimodal baselines such as FedInMM and FedMAC. This reduction in communication directly translates into faster communication-bound rounds in bandwidth-constrained environments.

The reported CPU and GPU usages primarily reflect the cost of batched tensor operations (forward and backward passes) in the modality encoders, multi-way transformer fusion, adaptive gating, projection heads, and loss computations, which are offloaded to the GPU, together with orchestration, PyTorch runtime overhead, data loading, and communication management handled by the CPU. RAM utilization captures the joint footprint of model parameters, intermediate activations, and small prototype buffers. As shown in Table 7, MODALIS exhibits GPU utilization of approximately 70% and RAM utilization of roughly 31% on the evaluation node, which is very close to multimodal FL methods with similarly rich architectures such as PmcmFL, FedMAC, FedInMM, and MFCPL. CPU utilization for MODALIS (approximately 44%) is likewise on par with these advanced baselines. In contrast, FedAvg and FedProto show somewhat lower GPU and RAM usage, but require significantly more communication per round. Overall, these results indicate that MODALIS targets moderately capable edge devices or edge servers, where the additional local computation relative to FedAvg or FedProto is on the same order as that of other state-of-the-art multimodal FL methods and is compensated by the substantial communication savings.

Finally, the drastically reduced communication payload, combined with moderate local resource usage, enables MODALIS to achieve competitive end-to-end convergence times. In the simulated, computation-intensive setting considered, MODALIS converges in approximately 4.5 hours, which is comparable to or faster than other advanced multimodal approaches such as FedInMM, FedMAC, PmcmFL, and MFCPL, while achieving the highest final Macro-F1 score of 83.4%. This behavior underscores that the prototype-centric design delivers a favorable trade-off

Table 6Cross-dataset hyperparameter transfer under moderate heterogeneity ($\beta = 0.5, p_m = 0.25$).

Method	Target dataset	Per-dataset tuned (%)	AVMNIST-tuned (%)	Δ (%)
FedAvg	CREMA-D	52.7 $\pm$ 0.6	49.7 $\pm$ 0.7	$\downarrow$ 5.7
	UCI-HAR	78.6 $\pm$ 0.5	75.4 $\pm$ 0.6	$\downarrow$ 4.1
	CRISIS-MMD	57.5 $\pm$ 0.7	54.0 $\pm$ 0.8	$\downarrow$ 6.1
MFCPL	CREMA-D	62.8 $\pm$ 0.4	59.2 $\pm$ 0.5	$\downarrow$ 3.6
	UCI-HAR	88.5 $\pm$ 0.3	86.5 $\pm$ 0.4	$\downarrow$ 2.0
	CRISIS-MMD	68.2 $\pm$ 0.4	64.7 $\pm$ 0.5	$\downarrow$ 3.5
MODALIS	CREMA-D	63.3 $\pm$ 0.4	62.1 $\pm$ 0.5	$\downarrow$ 1.9
	UCI-HAR	89.3 $\pm$ 0.3	88.2 $\pm$ 0.4	$\downarrow$ 1.2
	CRISIS-MMD	68.7 $\pm$ 0.4	67.4 $\pm$ 0.5	$\downarrow$ 1.9

Table 7Resource cost and training time using AVMNIST ($\beta = 0.1, p_m = 0.5$).

Method	Communication Cost (MB) ^a	Training Time (min) ^a	Total Time (hr)	CPU (%)	GPU (%)	RAM (%)	F1 score (%)
Centralized	N/A	N/A	$\approx$ 1.9	$\approx$ 52	$\approx$ 78	$\approx$ 34	87.9 $\pm$ 0.2
FedAvg	$\approx$ 4.75	$\approx$ 2.03	$\approx$ 6.1	$\approx$ 37	$\approx$ 54	$\approx$ 26	69.5 $\pm$ 0.9
FedProto	$\approx$ 0.03	$\approx$ 1.12	$\approx$ 3.7	$\approx$ 33	$\approx$ 50	$\approx$ 24	74.8 $\pm$ 0.6
FedInMM	$\approx$ 4.85	$\approx$ 1.43	$\approx$ 4.8	$\approx$ 41	$\approx$ 67	$\approx$ 29	79.7 $\pm$ 0.5
CreamFL	$\approx$ 4.75	$\approx$ 1.45	$\approx$ 4.8	$\approx$ 39	$\approx$ 64	$\approx$ 28	78.3 $\pm$ 0.6
PmcmFL	$\approx$ 4.80	$\approx$ 1.42	$\approx$ 4.7	$\approx$ 43	$\approx$ 69	$\approx$ 30	80.3 $\pm$ 0.4
FedMAC	$\approx$ 4.85	$\approx$ 1.44	$\approx$ 4.8	$\approx$ 42	$\approx$ 68	$\approx$ 30	79.9 $\pm$ 0.5
MFCPL	$\approx$ 0.27	$\approx$ 1.46	$\approx$ 4.6	$\approx$ 41	$\approx$ 66	$\approx$ 29	80.0 $\pm$ 0.5
MODALIS (our work)	$\approx$ 0.11	$\approx$ 1.35	$\approx$ 4.5	$\approx$ 44	$\approx$ 70	$\approx$ 31	83.4 $\pm$ 0.4

^a The results are reported as the average uplink payload per client message in each communication round.

between computational efficiency, communication cost, and model performance.

6.5. Ablation studies

To gain deeper insight into the behavior of MODALIS, this section presents a set of ablation and sensitivity studies focusing on: (i) the main architectural components and loss terms; (ii) the impact of the underlying communication topology; and (iii) the sensitivity of key hyperparameters governing the multi-objective loss, prototype updates, and embedding dimensionality, with the explicit goal of quantifying how robust MODALIS is to moderate changes in these settings and whether extensive per-scenario fine-tuning is required. Unless otherwise stated, all experiments are conducted on the AVMNIST dataset under moderate heterogeneity ($\beta = 0.5, p_m = 0.25$).

Ablation study of MODALIS components. The first study evaluates the contributions of the core architectural blocks and each term in the multi-objective loss of Eq. (4). Table 8 reports the resulting average Macro-F1 scores and relative computational overhead for variants in which a single component is removed or modified at a time.

Removing the Contextual Null Embedding module ($\mathcal{N}_M$) resulted in an absolute F1 score reduction of 47.1%, confirming that naive imputation of missing modalities severely harms performance even when all other components are kept unchanged. Eliminating the gating module ($\mathcal{G}$) or masked fusion $\mathcal{F}$ similarly results in large drops, indicating that both adaptive modality weighting and explicit masking of imputed or absent modalities are necessary to obtain robust multimodal representations. The adaptive EMA update for prototypes also plays an important role: replacing it with a fixed momentum degrades accuracy while slightly reducing per-round computation, showing that the small added complexity of the adaptive scheme yields a favorable accuracy-efficiency trade-off. Ablations on the loss function further reveal that

both $\mathcal{L}_{\text{FAL}}$ and $\mathcal{L}_{\text{DCL}}$ are critical under heterogeneous data, resulting in absolute drops of 28.1% and 39.8%, respectively. In addition, $\mathcal{L}_{\text{MCL}}$ and $\mathcal{L}_{\text{PR}}$ serve as lighter regularizers, providing consistent yet more modest improvements. Turning off the core prototype exchange mechanism and forcing clients to train in isolation with their prototypes resulted in a 35.1% drop, confirming that collaborative learning through prototype sharing is the cornerstone of success. These results strongly justify the comprehensive design of MODALIS, as the severe degradation in model performance vastly outweighs the minimal computational time savings gained from removing any single component. The study also validated the choice of a single local epoch ($T = 1$), as increasing T to five local epochs significantly harmed performance due to client drift, while drastically increasing computation time.

Topology sensitivity. To assess robustness with respect to the communication topology, MODALIS was evaluated under several decentralized graphs with the same number of clients. Table 9 compares the default static ER graph with edge probability $p_{ER} = 0.9$ to: (i) a fully connected graph; (ii) a dynamic ER graph with $p_{ER} = 0.9$ in which neighbors are resampled randomly at each round; (iii) a sparser static ER graph with $p_{ER} = 0.3$; (iv) a sparse dynamic ER graph with $p_{ER} = 0.3$; and (v) a ring topology with two neighbors per client.

Fully connected and dense ER graphs achieve the highest scores, as expected: the fully connected topology offers a small improvement to 90.5%, and dynamic ER with $p_{ER} = 0.9$ reaches 90.3% relative to the static ER baseline. Both static and dynamic sparse ER topologies incur only moderate degradation, with Macro-F1 dropping to 88.2% and 88.5%, despite their substantially lower average degree. The ring topology exhibits the largest drop (a 5.0-point decrease), reflecting the slower information mixing inherent to very sparse, low-diameter graphs. Importantly, the dynamic ER graphs perform comparably to their static counterparts at the same density, indicating that the prototype-based aggregation mechanism is robust to moderate topology dynamics and

Table 8Ablation study of MODALIS components using AVMNIST dataset ($\beta = 0.5, p_m = 0.25$).

Variant configuration	Macro-F1 (%)	Time per round (s)
MODALIS (full model)	90.1±0.2	81.0
<i>Architectural components</i>		
w/o Contextual Null Embed ($\mathcal{N}_M$)	43.0±1.5 ($\downarrow 47.1$)	78.3 ($\downarrow 3.3\%$)
w/o Gating Module ($\mathcal{G}$)	68.0±0.9 ($\downarrow 22.1$)	79.6 ($\downarrow 1.7\%$)
w/o Masked Fusion in $\mathcal{F}$	55.0±1.1 ($\downarrow 35.1$)	77.1 ($\downarrow 4.8\%$)
w/o Adaptive EMA (fixed $m_t = 0.9$)	79.0±0.7 ($\downarrow 11.1$)	72.9 ($\downarrow 10.0\%$)
<i>Loss components from Eq. (4)</i>		
w/o $\mathcal{L}_{\text{FAL}}$	62.0±1.0 ($\downarrow 28.1$)	78.3 ($\downarrow 3.3\%$)
w/o $\mathcal{L}_{\text{DCL}}$	50.3±1.2 ($\downarrow 39.8$)	74.3 ($\downarrow 8.3\%$)
w/o $\mathcal{L}_{\text{MCL}}$	87.4±0.4 ($\downarrow 2.7$)	79.6 ($\downarrow 1.7\%$)
w/o $\mathcal{L}_{\text{PR}}$	89.2±0.3 ($\downarrow 0.9$)	79.6 ($\downarrow 1.7\%$)
<i>Protocol variants</i>		
$T = 5$ local epochs per round	82.2±0.7 ($\downarrow 7.9$)	340.2 ($\uparrow 320.0$)
Local prototypes only (no exchange)	55.0±1.3 ($\downarrow 35.1$)	64.8 ($\downarrow 20.0\%$)

Table 9Impact of network topology on MODALIS using AVMNIST ($\beta = 0.5, p_m = 0.25$).

Topology configuration	Macro-F1 (%)	Time per round (s)
Static ER ($p_{\text{ER}} = 0.9$)	90.1±0.2	81.0
Fully connected	90.5±0.2 ($\uparrow 0.4$)	82.5
Dynamic ER ($p_{\text{ER}} = 0.9$)	90.3±0.3 ($\uparrow 0.2$)	81.4
Sparse static ER ($p_{\text{ER}} = 0.3$)	88.2±0.5 ($\downarrow 1.9$)	79.2
Sparse dynamic ER ($p_{\text{ER}} = 0.3$)	88.5±0.4 ($\downarrow 1.6$)	79.8
Ring (2 neighbors per client)	85.1±0.6 ($\downarrow 5.0$)	78.6

does not rely on a fixed neighbor set. These trends empirically support the theoretical dependence on the mixing properties of the communication matrix captured by ρ in [Assumption 4](#). When scaling to large N , preserving a small-world or expander-like structure with bounded average degree keeps the spectral gap large and the effective diameter small, so prototype information continues to propagate efficiently even with hundreds or thousands of clients. In contrast, very sparse, chain-like topologies have diameters that grow linearly with N , and thus require substantially more communication rounds or additional mechanisms (e.g., hierarchical aggregation or occasional long-range links) to reach a comparable level of prototype consensus.

Hyperparameter sensitivity and representation capacity. A complementary sensitivity analysis investigates the hyperparameters that most directly control prototype learning and the multi-objective loss in [Eq. \(4\)](#). [Table 10](#) reports the effect of varying: (i) the weights of the alignment and contrastive losses ($w_{\text{FAL}}, w_{\text{DCL}}$); (ii) the bounds of the adaptive EMA momentum ($m_{\min}, m_{\max}$); (iii) the $\mathcal{L}_{\text{DCL}}$ parameters (τ, μ, s); and (iv) the embedding dimensionality d of the shared representation and prototype space. In all cases, differences are reported relative to the common baseline configuration ($w_{\text{FAL}}=0.5, w_{\text{DCL}}=0.2, (m_{\min}, m_{\max})=(0.5, 0.99), (\tau, \mu, s)=(0.1, 0.3, 30), d=128$).

The results exhibit a broad and smooth plateau around the default configuration. For w_{FAL} , setting the weight to zero reduces Macro-F1 from 90.1% to 86.8%, and a small weight of 0.1 yields 87.5%. Increasing w_{FAL} to 1.0 slightly reduces performance to 89.8%, indicating that the method is not overly sensitive to moderate over-weighting as long as the task loss remains dominant. A similar pattern holds for w_{DCL} : removing the term altogether reduces Macro-F1 to 86.9%, while values in the range [0.05, 0.5] keep performance within 1.2 points of the baseline.

Table 10Sensitivity of MODALIS to key hyperparameters on AVMNIST ($\beta = 0.5, p_m = 0.25$).

Group	Setting	Macro-F1 (%)
Baseline	—	90.1±0.2
Loss weight w_{FAL}	$w_{\text{FAL}} = 0.0$	86.8±0.5 ($\downarrow 3.3$)
	$w_{\text{FAL}} = 0.1$	87.5±0.4 ($\downarrow 2.6$)
	$w_{\text{FAL}} = 0.5$	90.1±0.2
	$w_{\text{FAL}} = 1.0$	89.8±0.3 ($\downarrow 0.3$)
Loss weight w_{DCL}	$w_{\text{DCL}} = 0.0$	86.9±0.5 ($\downarrow 3.2$)
	$w_{\text{DCL}} = 0.05$	88.9±0.4 ($\downarrow 1.2$)
	$w_{\text{DCL}} = 0.2$	90.1±0.2
	$w_{\text{DCL}} = 0.5$	89.5±0.3 ($\downarrow 0.6$)
EMA bounds ($m_{\min}, m_{\max}$)	(0.1, 0.5)	88.1±0.5 ($\downarrow 2.0$)
	(0.3, 0.9)	89.4±0.3 ($\downarrow 0.7$)
	(0.5, 0.99)	90.1±0.2
	(0.7, 0.995)	89.2±0.4 ($\downarrow 0.9$)
DCL params (τ, μ, s)	(0.9, 0.999)	88.9±0.4 ($\downarrow 1.2$)
	(0.1, 0.3, 30)	90.1±0.2
	(0.07, 0.3, 30)	89.7±0.3 ($\downarrow 0.4$)
	(0.2, 0.3, 30)	89.1±0.3 ($\downarrow 1.0$)
Embedding dim. d	(0.1, 0.0, 30)	86.4±0.6 ($\downarrow 3.7$)
	(0.1, 0.5, 30)	89.5±0.4 ($\downarrow 0.6$)
	(0.1, 0.3, 50)	88.8±0.4 ($\downarrow 1.3$)
	$d = 64$	88.6±0.3 ($\downarrow 1.5$)
Embedding dim. d	$d = 96$	89.5±0.3 ($\downarrow 0.6$)
	$d = 128$	90.1±0.2
	$d = 256$	90.4±0.3 ($\uparrow 0.3$)

The EMA bounds ($m_{\min}, m_{\max}$) show a gradual effect: very conservative updates (0.1, 0.5) reduce Macro-F1 to 88.1%, while slightly tighter or looser ranges such as (0.3, 0.9) or (0.7, 0.995) remain within 0.7–0.9 points of the baseline. Extremely slow adaptation (0.9, 0.999) results in a drop to 88.9%. For the DCL parameters, removing the angular margin ($\mu = 0$) is particularly detrimental, decreasing Macro-F1 from 90.1% to 86.4%. Changing the temperature from 0.1 to 0.2, or slightly adjusting μ or s , results in much smaller drops (between 0.4 and 1.3 points), indicating

that the contrastive term is robust to moderate perturbations around the default.

Finally, the representation capacity experiments show that increasing the embedding dimension d from 64 to 128 improves Macro-F1 from 88.6% to 90.1%, whereas further increasing d to 256 yields only a marginal additional improvement to 90.4% at the cost of doubling the prototype dimensionality and hence the communication payload. This behavior is consistent with the communication-cost analysis in Lemma 1, which shows that beyond a certain point, additional embedding capacity yields diminishing returns relative to the linear growth in prototype size

7. Discussion

The empirical results presented in Section 6 strongly support the efficacy of MODALIS. This section contextualizes these findings by highlighting the framework's core contributions relative to the state of the art, analyzing its key trade-offs, and candidly addressing its limitations, thereby outlining clear directions for future research.

Novelty and core contributions: The primary contribution of MODALIS is the synthesis of a fully decentralized communication protocol with a sophisticated, multimodal-aware client architecture. Unlike prior art, which typically addresses decentralization and multimodal heterogeneity in isolation, MODALIS tackles both simultaneously. Its prototype-centric protocol directly circumvents the communication bottlenecks and knowledge dilution issues inherent to parameter-averaging DFL methods. Architecturally, the framework's novelty lies in its end-to-end approach to handling incomplete data. The Contextual Null Embedding module represents a significant design advance over static imputation, enabling intelligent, feature-conditioned generation of missing modality representations. This robust input processing is then complemented by an adaptive gating mechanism that dynamically weighs the importance of each modality, followed by a masked multi-way transformer for fusion, which explicitly prevents less reliable imputed data from corrupting the final representation. Furthermore, this entire process is guided by a multi-objective loss function that provides a principled mechanism for enforcing representation consistency, not only across the decentralized network ($\mathcal{L}_{\text{FAL}}$, $\mathcal{L}_{\text{DCL}}$) but also across modalities within a single client ($\mathcal{L}_{\text{MCL}}$). This synergy is what enables effective cross-modal knowledge transfer, a capability validated by the strong performance of unimodal clients (see Table 5).

Interpreting performance under heterogeneity: The resilience of MODALIS is most evident in high-stress scenarios ($\beta = 0.1$). As shown in Table 4, while the performance of all methods degrades with increasing heterogeneity, MODALIS consistently maintains the highest F1 scores. For example, on AVMNIST with $p_m = 0.5$, MODALIS achieves an F1 score of 83.4%, a significant +3.1% absolute gain over the next-best federated method, PmcmFL, and a 13.9% gain over the standard FedAvg. This superior performance can be directly attributed to the prototype-guided multi-objective loss function. By regularizing local training with aggregated prototypes from neighbors (via losses such as $\mathcal{L}_{\text{FAL}}$ and $\mathcal{L}_{\text{DCL}}$), MODALIS effectively counteracts client drift induced by local data biases, enabling the network to converge to a more robust and generalized solution. Furthermore, the outstanding performance of unimodal clients (Table 5) confirms the effectiveness of the cross-modal knowledge transfer mechanism. A client possessing only image data, for example, successfully leverages audio prototypes from its neighbors to regularize its training, a feat made possible only by the coherent cross-modal representations enforced by the framework's design.

Efficiency and decentralization trade-offs: The MODALIS prototype-exchange mechanism yields substantial communication savings, estimated at over $40\times$ compared to full model exchange methods (Table 7), which is a critical advantage for deployment in bandwidth-constrained environments. A noteworthy comparison is with FedProto, a centralized prototype-based method that is even more communication-efficient. The consistent underperformance of

FedProto relative to MODALIS highlights a key trade-off: MODALIS leverages decentralized, peer-to-peer communication to achieve a higher-quality consensus on prototypes, resulting in better accuracy with only a modest increase in communication overhead compared to server-centric prototype approaches.

Communication, computation, and privacy: From a system-design perspective, the prototype-centric protocol of MODALIS scales gracefully with the number of clients, since each device exchanges a fixed-size prototype library only with its neighbors, so the per-round communication cost scales with the node degree rather than the total network size. The convergence analysis in Section 5 makes this dependence explicit through the mixing parameter ρ . Moreover, the empirical study in Sections 6.4 and 6.5 shows that MODALIS maintains competitive wall-clock training time and moderate resource usage compared to recent multimodal FL baselines, despite its richer architecture and fusion modules. At the same time, Section 5.3 clarifies that the current design offers only intuitive privacy advantages over gradient sharing and does not provide formal guarantees, such as those provided by DP. In particular, clients with very few samples per class are more vulnerable to membership inference and related attacks, because their class prototypes are averaged over a small number of examples and may encode more sample-specific features.

Topology and convergence: Scaling to much larger networks (e.g., $N \geq 10^3$) primarily affects the number of rounds required for prototypes to mix globally, rather than the local computation or per-round communication budget of each client. For well-connected small-world or expander-like topologies, the graph diameter and mixing time grow only logarithmically with N , so the spectral gap and, consequently, the parameter ρ remain favorable. In this regime, prototype convergence degrades slowly as the network grows, at the cost of a modest increase in communication rounds. In contrast, very sparse or elongated graphs have large diameters and poorer mixing properties, which translate into larger consensus-error terms in Theorem 1 and empirically slower propagation of information across distant regions of the network, as confirmed by the topology-sensitivity study in Section 6.5 comparing dense ER graphs, sparser graphs, and a ring topology. For large deployments, degree-capped small-world overlays and hierarchical or clustered communication backbones can keep node degree and per-round communication bounded while preserving a small effective diameter and fast prototype convergence. For resource-constrained devices, lightweight variants of MODALIS with smaller backbones, lower-dimensional embeddings, or pruned auxiliary losses are a promising direction for low-power federated multimodal learning.

Limitations and Scope: Despite its strong performance, MODALIS has several limitations. First, the present study explores sparser and dynamic ER topologies. However, the experiments still assume a fixed set of participating clients and abstract away explicit client churn, physical node mobility, and realistic network latencies. This design isolates the impact of the proposed multimodal prototype-learning mechanisms from confounding networking effects. Moreover, while the neighbor-to-neighbor communication model scales favorably in terms of per-round cost, in extremely large or very sparsely connected networks, the number of rounds required to achieve a given level of global consensus may increase due to slower information mixing. In many real-world decentralized systems, such as mobile ad-hoc networks, clients may join or leave at any time, and links can become unreliable. Therefore, extending the framework to support dynamic topologies is an essential next step. Second, as noted in Section 5, prototype sharing offers stronger privacy than gradient sharing but falls short of formal guarantees, such as differential privacy. Adding rigorous privacy mechanisms will be necessary for applications that handle highly sensitive data. Third, the cross-dataset transfer analysis in Section 6.3.4 is intentionally limited to configuration-level transfer, where a single set of hyperparameters and architectural choices tuned on a source dataset is reused when training on other datasets. Fully zero-shot deployment of a model trained end-to-end on one dataset directly to another with a different label or

modality structure is not addressed in the current work, and more advanced forms of cross-domain adaptation, such as feature-level transfer or fine-tuning under distribution shift, remain an important direction for future research. Fourth, although the experimental protocol already includes challenging configurations with Dirichlet parameter $\beta = 0.1$ and missing-modality proportion $p_m = 0.5$, even more extreme regimes are expected to induce stronger degradation of representation quality and downstream performance. Addressing such pathological settings may require additional mechanisms, such as adaptive client selection, curriculum-style training schedules, or task-aware regularization, which lie beyond the scope of the present study and represent an important avenue for future work.

8. Conclusion

This paper presented MODALIS, a DFL framework engineered to optimize collaborative training on heterogeneous multimodal data, delivering significant performance gains while drastically reducing network consumption. By combining a communication-efficient prototype-exchange protocol with a sophisticated client-side architecture and a multi-objective loss function, MODALIS enables effective collaborative learning without a central server. The core innovations, including contextual null embeddings for missing modalities and prototype-guided alignment losses, were shown to be critical for achieving state-of-the-art performance. Empirical results across four datasets confirmed that MODALIS establishes a new benchmark for decentralized multimodal intelligence. It significantly outperforms existing methods by improving F1 scores by up to 4% under severe data heterogeneity, while simultaneously reducing communication costs by over 40x compared to standard parameter-averaging approaches.

Future research will proceed in several key directions. The primary focus will be on extending the framework to accommodate dynamic network topologies, enabling it to handle client churn and evolving connectivity. Another crucial avenue is to incorporate formal privacy guarantees and robustness to adversarial attacks, making the framework suitable for deployments with strict security requirements. Finally, investigating methods to enhance scalability for scenarios with extremely large numbers of classes or modalities and exploring the integration of fairness-aware mechanisms to ensure equitable performance will be important for broadening the practical impact of this work.

CRedit authorship contribution statement

Enrique Tomás Martínez Beltrán: Writing - original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **G  r  me Bovet:** Resources, Project administration, Conceptualization; **Gregorio Mart  nez P  rez:** Supervision, Resources, Project administration, Funding acquisition; **Alberto Huertas Celdr  n:** Writing - review & editing, Validation, Supervision, Methodology, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Enrique Tomas Martinez Beltran reports financial support was provided by Fundaci  n S  neca, Regi  n de Murcia (Spain). Gregorio Martinez Perez reports financial support was provided by the Swiss Federal Office for Defense Procurement (armasuisse). Gregorio Martinez Perez reports financial support was provided by the European Defence Fund (EDF). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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